Practical: 1

AIM: Write a HTML page to print Hello World in bold and italic font.

CODE:

```
<html>

<head>

<title> Practical 1 </title>

</head>

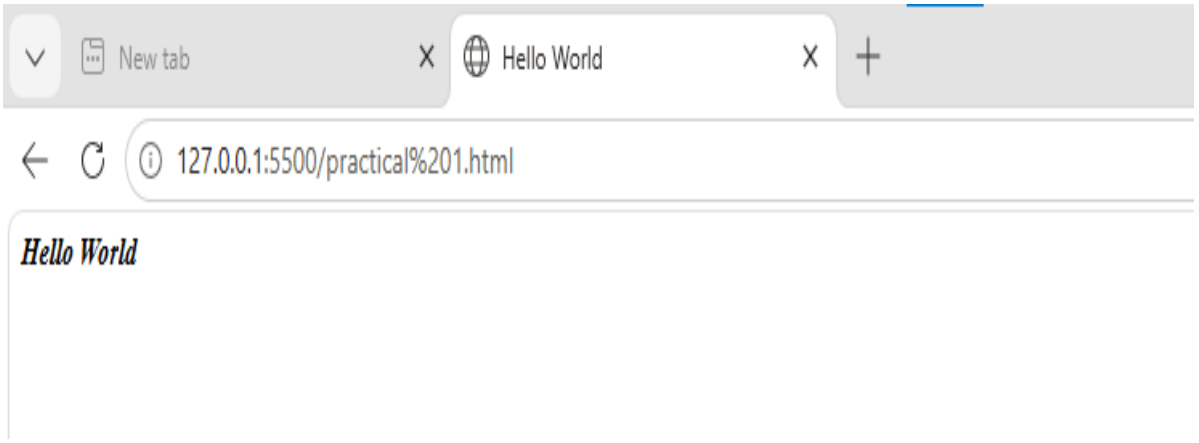
<body>

<b><i>Hello World</i></b>

</body>

</html>
```

OUTPUT:



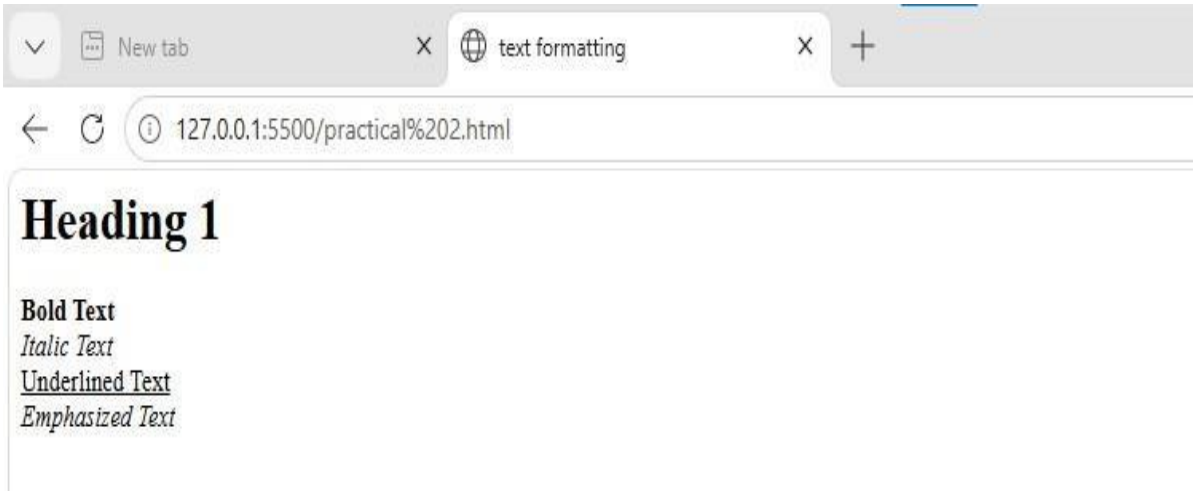
Practical: 2

AIM: Display various text formatting methods available in HTML.(i.e.<h1>,<b>,<u> etc…)

CODE:

```
<html>
  <head>
    <title>text formatting</title>
  </head>
  <body>
    <h1>Heading 1</h1>
    <b>Bold Text</b><br>
    <i>Italic Text</i><br>
    <u>Underlined Text</u><br>
    <em>Emphasized Text</em>
  </body>
</html>
```

OUTPUT:



Practical: 3

AIM: Create a HTML file using special characters.

CODE:

```
<html>
<head>
<title>PR_3.</title>
</head>
<body>
<h1>special character</h1>
<p>
    &lt;HTML&gt; is a language &amp;
    it is used to create web pages.
    &copy; 2026 &reg;
</p>
<p>
    &lt; &quot;Geopolitics&quot; &copy;
   studies how countries interact with each other. Nations use trade &amp;
   diplomacy to protect their interests. Power &divide;
   resources can influence global relations. &reg; &gt;
</p>
<p>
   &lt; &quot;External Affairs&quot; &copy; deals with relations between countries.
   Nations use trade &amp; diplomacy to build strong relationships.
   Global cooperation &divide; conflicts can affect international relations.
   Countries spend &pound; billions on foreign policies. &reg; &gt;
</p>
</body>
</html>
```

OUTPUT:



special character

<HTML> is a language & it is used to create web pages. © 2026 ®

< "Geopolitics" © studies how countries interact with each other. Nations use trade & diplomacy to protect their interests. Power ÷ resources can influence global relations. ® >

< "External Affairs" © deals with relations between countries. Nations use trade & diplomacy to build strong relationships. Global cooperation ÷ conflicts can affect international relations. Countries spend £ billions on foreign policies. ® >

Practical 4

AIM: Create a HTML file which displays 3 images at LEFT, RIGHT and CENTER respectively in the browser.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Pr_4</title>
</head>

<body>
<h1>Conor McGregor</h1>

   <img src="photo.jpg" align="left" alt="Conor McGregor" width="100"
   height="100">
   <p>
      <b>Conor McGregor is an Irish professional mixed martial artist
      known for his striking and success in the UFC.</b>
   </p>

   <br><br><br><br><br>

   <img src="photo.jpg" align="right" alt="Conor McGregor" width="100"
    height="100">
   <p>
      <b>He became the first UFC fighter to hold titles in two weight
      divisions at the same time.</b>
   </p>

   <br><br><br><br><br>

   <center>
      <img src="photo.jpg" alt="Conor McGregor" width="100" height="100">
      <p>
         <b>McGregor is one of the most famous and successful fighters
         in the history of mixed martial arts.</b>
      </p>
   </center>

</body>
</html>
```

OUTPUT:

Conor McGregor



Conor McGregor is an Irish professional mixed martial artist known for his striking and success in the UFC.

He became the first UFC fighter to hold titles in two weight divisions at the same time.





McGregor is one of the most famous and successful fighters in the history of mixed martial arts.

Practical: 5

AIM: Create a HTML file which contains hyperlinks.

CODE:

```
<!DOCTYPE html>

<html>

  <head>

    <title>link</title>

  </head>

  <body>

    <h2>HTML Hyperlinks</h2>

    <a href="practical 2.html">practical 2</a>

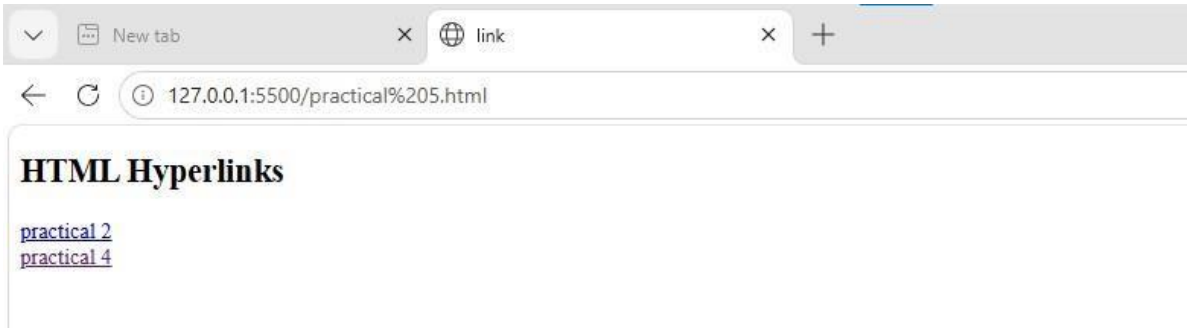
    <br>

    <a href="practical 4.html">practical 4</a>

  </body>

</html>
```

OUTPUT:



Conor McGregor



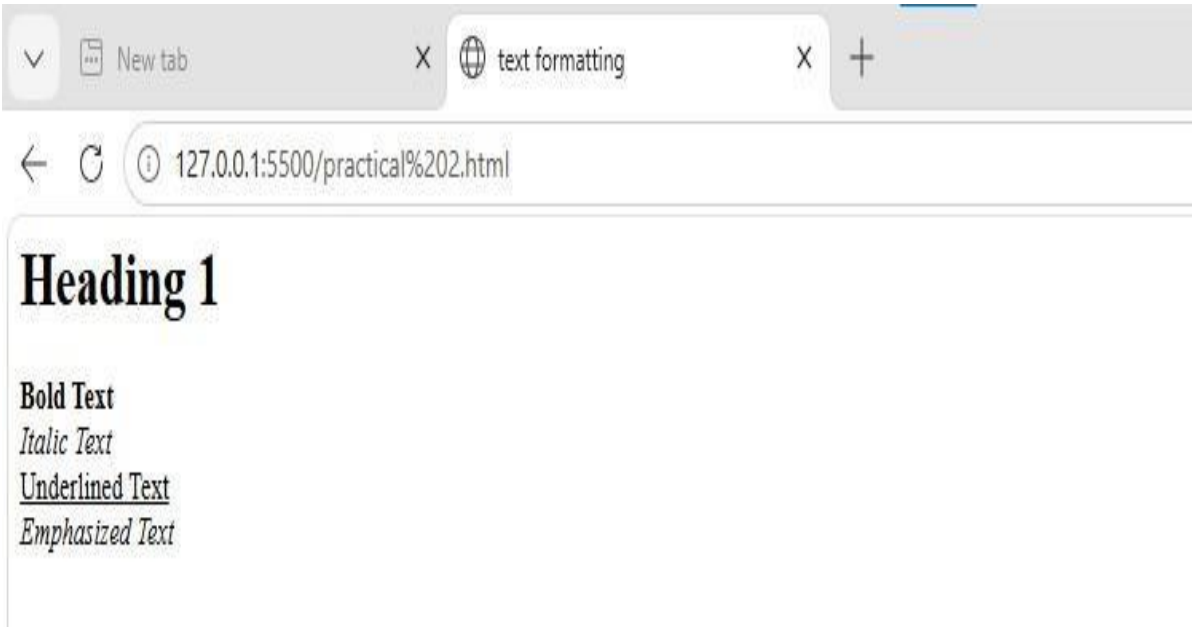
Conor McGregor is an Irish professional mixed martial artist known for his striking and success in the UFC.

He became the first UFC fighter to hold titles in two weight divisions at the same time.





McGregor is one of the most famous and successful fighters in the history of mixed martial arts.



Practical: 6

AIM: Table of Contents Chapter

1: Introduction Chapter

2: What is HTML? Chapter

3: What is Javascript?.....By clicking on the link takes to the respective topic within the same page.

CODE:

```
<!DOCTYPE html>
<html>
<head>
<title>Table of Contents</title>
</head>
<body>
<h2>Table of Contents</h2>
<a href="#intro">Chapter 1: Introduction</a><br>
<a href="#html">Chapter 2: What is HTML?</a><br>
<a href="#js">Chapter 3: What is JavaScript?</a><br>
<hr>
<h2 id="intro">Chapter 1: Introduction</h2>
<p>
HTML (HyperText Markup Language) is used to create and structure web pages.
It uses tags to add headings, paragraphs, images, links, tables, and forms.
HTML provides the basic structure of a website and works with CSS and
JavaScript.

<br>JavaScript is a programming language used to make web pages interactive
and dynamic. It is used for form validation, animations, pop-up messages,
and updating content without reloading the page.

<br>HTML5 is the modern version of HTML and supports multimedia, forms,
semantic elements, and better web applications. HTML provides the structure,
```

CSS controls the design, and JavaScript adds functionality to a webpage.

<hr>
<h2 id="html">Chapter 2: What is HTML?</h2>

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. It was developed by Tim Berners-Lee in 1991. HTML uses tags and elements to organize content such as headings, paragraphs, images, links, tables, lists, and forms. Web browsers read these tags and display the content in a structured format.

HTML5 is the modern version of HTML and provides features such as multimedia, semantic elements, improved forms, and better support for web applications. HTML is easy to learn, platform-independent, and supported by all modern web browsers. HTML provides the basic structure of a webpage, while CSS is used for styling and JavaScript adds interactivity. Together, HTML, CSS, and JavaScript are the main technologies used in modern web development.

<hr>
<h2 id="js">Chapter 3: What is JavaScript?</h2>

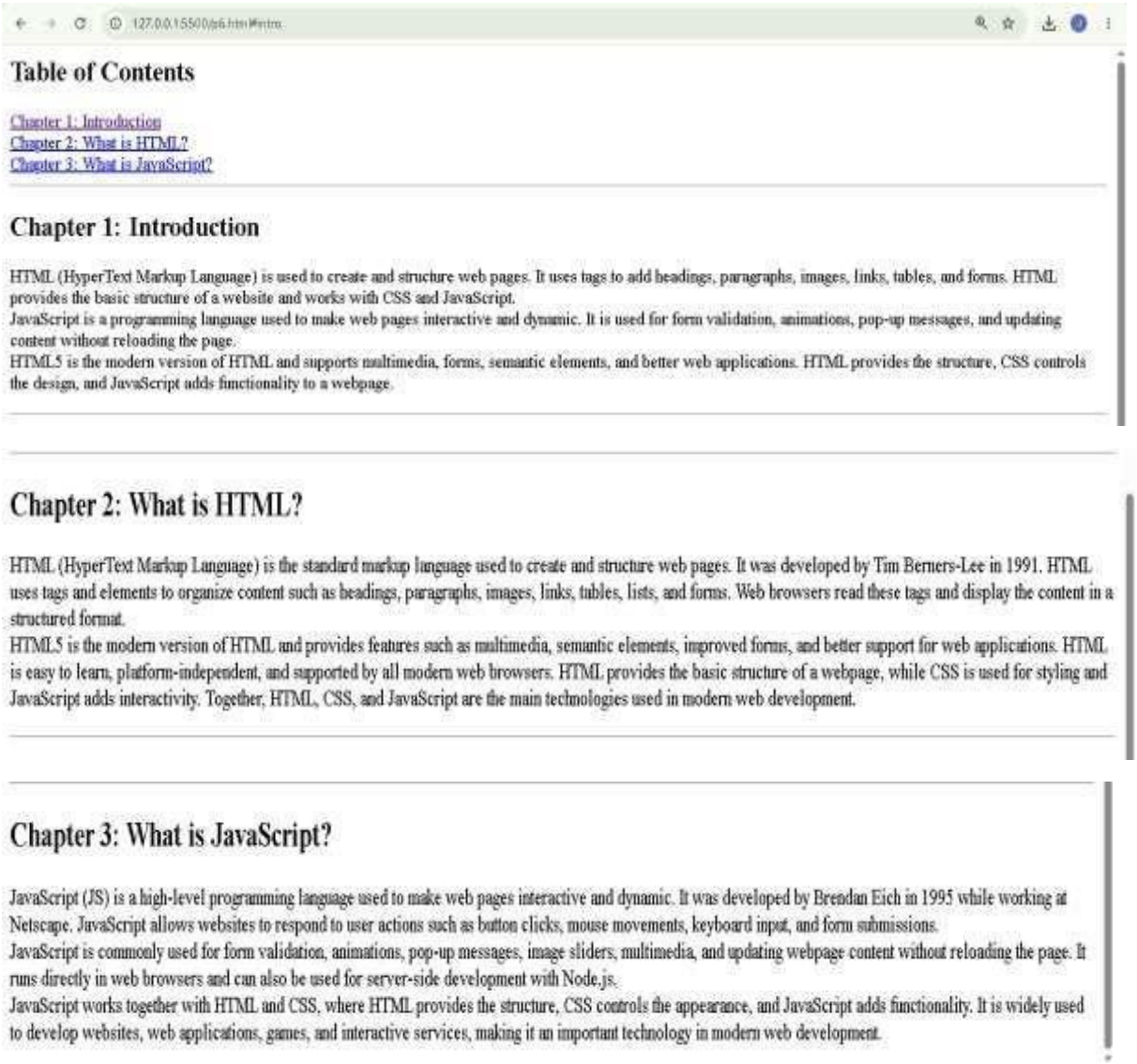
JavaScript (JS) is a high-level programming language used to make web pages interactive and dynamic. It was developed by Brendan Eich in 1995 while working at Netscape. JavaScript allows websites to respond to user actions such as button clicks, mouse movements, keyboard input, and form submissions.

JavaScript is commonly used for form validation, animations, pop-up messages, image sliders, multimedia, and updating webpage content without

reloading the page. It runs directly in web browsers and can also be used for server-side development with Node.js.

JavaScript works together with HTML and CSS, where HTML provides the structure, CSS controls the appearance, and JavaScript adds functionality.
It is widely used to develop websites, web applications, games, and interactive services, making it an important technology in modern web development.
</p>
</body>
</html>

OUTPUT:



Practical: 7

AIM: Create a HTML page as given below List of Subjects

1. Computer Engineering Department

a. Software Engineering

b. Information Security

c. Computer Graphics

2. Electrical Engineering Department

- **Electrical Machine**
- **Power Electronics**
- **Micro Controller**

3. Computer Engineering Is a discipline that integrates several fields required to develop computer systems.

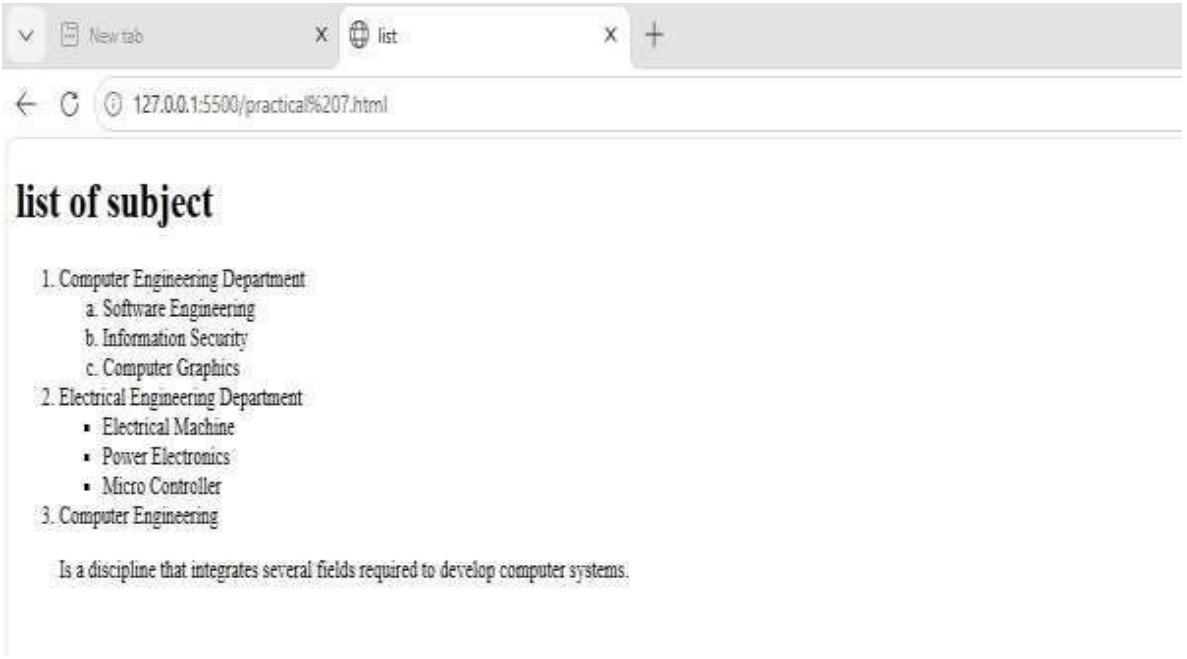
CODE:

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      list
    </title>
  </head>
  <body>
    <h1>
      list of subject
    </h1>
    <ol>
      <li>
        Computer Engineering Department
      </li>
      <ol type="a">
        <li>
```

```
            Software Engineering
          </li>
          <li>
            Information Security
          </li>
          <li>
            Computer Graphics
          </li>
        </ol>
        <li>
          Electrical Engineering Department
        </li>
        <ul type="square">
          <li>
            Electrical Machine
          </li>
          <li>
            Power Electronics
          </li>
          <li>
           Micro Controller
          </li>
        </ul>
        <li>
          Computer Engineering
        </li>
      </ol><dl>
        <dt>
          <dd>Is a discipline that integrates several fields required to develop computer
systems.</dd>
        </dt>
```

```
    </dl>
  </body>
</html>
```

OUTPUT:



Practical: 8

AIM: Create table with ROWSPAN and COLSPAN attribute of TABLE in HTML (Prepare timetable of your class).Include CELLSPACING & CELL PADDING.

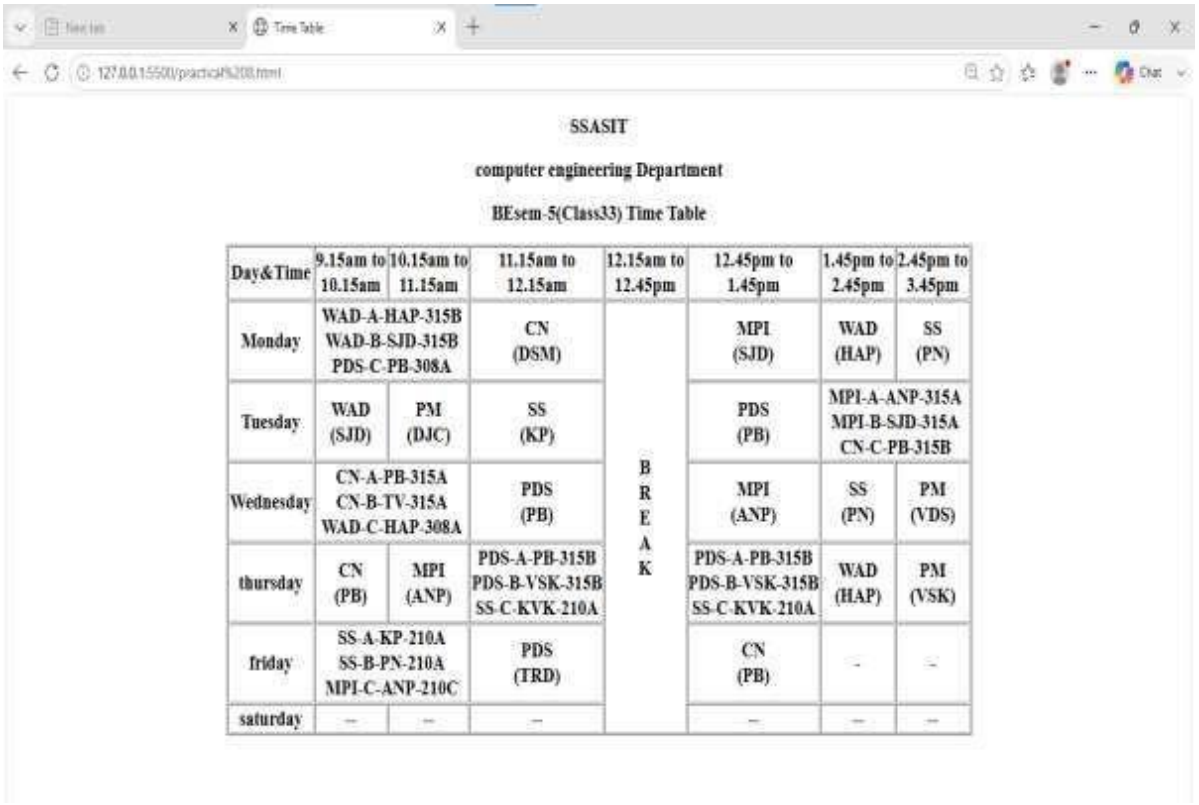
CODE:

```
<!DOCTYPE html>
<html>
  <head>
    <title>
      Time Table
    </title>
  </head>
  <body>
    <h2 align="center">SSASIT</h2>
    <h2 align="center">computer engineering Department</h2>
    <h2 align="center">BEsem-5(Class33) Time Table<h2>
    <table border="1" align="center">
      <thead>
       <tr>
         <th>Day&Time</th>
         <th>9.15am to<br> 10.15am</th>
         <th>10.15am to<br> 11.15am</th>
         <th>11.15am to<br> 12.15am</th>
         <th>12.15am to<br> 12.45pm</th>
         <th>12.45pm to<br> 1.45pm</th>
         <th>1.45pm to<br> 2.45pm</th>
         <th>2.45pm to<br> 3.45pm</th>
       </tr>
      </thead>
      <tbody>
        <tr align="center">
          <td>Monday</td>
```

```
            <td  colspan="2">WAD-A-HAP-315B<br>WAD-B-SJD-315B<br>PDS-C-PB-308A
</td>
            <td>CN<br>(DSM)</td>
            <td rowspan="6">B<br>R<br>E<br>A<br>K</td>
            <td>MPI<br>(SJD)</td>
            <td>WAD<br>(HAP)</td>
            <td>SS<br>(PN)</td>
          </tr>
          <tr align="center">
            <td>Tuesday</td>
            <td>WAD<BR>(SJD)</td>
            <td>PM<br>(DJC)</td>
            <td>SS<br>(KP)</td>
            <td>PDS<br>(PB)</td>
            <td  colspan="8">MPI-A-ANP-315A<br>MPI-B-SJD-315A<br>CN-C-PB-315B</td>
          </tr>
          <tr align="center">
            <td>Wednesday</td>
            <td  colspan="2">CN-A-PB-315A<br>CN-B-TV-315A<br>WAD-C-HAP-308A</td>
            <td>PDS<br>(PB)</td>
            <td>MPI<br>(ANP)</td>
            <td>SS<br>(PN)</td>
            <td>PM<br>(VDS)</td>
          </tr>
          <tr align="center">
            <td>thursday</td>
            <td>CN<br>(PB)</td>
            <td>MPI<br>(ANP)</td>
            <td>PDS-A-PB-315B<br>PDS-B-VSK-315B<br>SS-C-KVK-210A</td>
            <td>PDS-A-PB-315B<br>PDS-B-VSK-315B<br>SS-C-KVK-210A</td>
            <td>WAD<br>(HAP)</td>
```

```
            <td>PM<br>(VSK)</td>
          </tr>
          <tr align="center">
            <td>friday</td>
            <td  colspan="2">SS-A-KP-210A<br>SS-B-PN-210A<br>MPI-C-ANP-210C</td>
            <td>PDS<br>(TRD)</td>
            <td>CN<br>(PB)</td>
            <td>-</td>
            <td>-</td>
          </tr>
          <tr align="center">
            <td>saturday</td>
            <td>--</td>
            <td>--</td>
            <td>--</td>
            <td>--</td>
            <td>--</td>
            <td>--</td>
          </tr>
        </tbody>
      </table>
    </body>
</html>
```

OUTPUT:



SSASIT

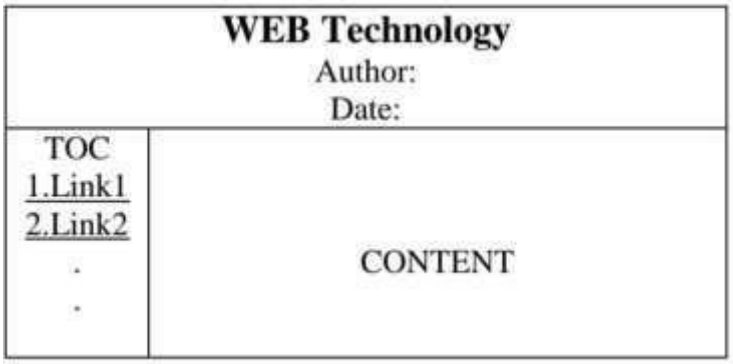
computer engineering Department

BEsem-5(Class33) Time Table

Day&Time	9.15am to 10.15am	10.15am to 11.15am	11.15am to 12.15am	12.15am to 12.45pm	12.45pm to 1.45pm	1.45pm to 2.45pm	2.45pm to 3.45pm
Monday	WAD-A-HAP-315B WAD-B-SJD-315B PDS-C-PB-308A		CN (DSM)		MPI (SJD)	WAD (HAP)	SS (PN)
Tuesday	WAD (SJD)	PM (DJC)	SS (KP)		PDS (PB)	MPI-A-ANP-315A MPI-B-SJD-315A CN-C-PB-315B	
Wednesday	CN-A-PB-315A CN-B-TV-315A WAD-C-HAP-308A		PDS (PB)		MPI (ANP)	SS (PN)	PM (VDS)
thursday	CN (PB)	MPI (ANP)	PDS-A-PB-315B PDS-B-VSK-315B SS-C-KVK-210A		PDS-A-PB-315B PDS-B-VSK-315B SS-C-KVK-210A	WAD (HAP)	PM (VSK)
friday	SS-A-KP-210A SS-B-PN-210A MPI-C-ANP-210C		PDS (TRD)		CN (PB)	-	-
saturday	–	–	–		–	–	–

Practical: 9

AIM: Create HTML page using Frames.



CODE:

1. **Index.html**

```
<html>
<head>
  <title>Frames</title>
</head>

  <frameset rows="25%,75%">

  <frame src="top.html" name="top">

  <frameset cols="25%,75%">

  <frame src="left.html" name="left">
  <frame src="content.html" name="content">
  </frameset>
  </frameset>
  </html>
```

2. **top.html**

```
<html>
<head>
  <title>Top</title>
</head>
```

```
<body align="center">
   <h3>WEB TECHNOLOGY</h3>
   Author : TARUNKUMAR JOSHI
   <br>
   Date : 18/08/2026
</body>
</html>
```

3. **left.html**

```
<html>
<head>
   <title>TOC</title>
</head>
<body>
   <b>TOC</b>
   <br><br>
<a href="introduction.html" target="content">
      1. Introduction
</a>
<br><br>
 <a href="content1.html" target="content">
      2. Contents
</a>
</body>
</html>
```

4. **content.html**

```
<html>
<head>
   <title>Content</title>
</head>
<body>
   <h2 align="center">CONTENT</h2>
```

```
</body>
</html>
```

5. **introduction.html**

```
<html>
<head>
  <title>Introduction</title>
</head>
<body>
  <h2>Introduction</h2>
  <p>
    HTML (HyperText Markup Language) is used to create and structure web pages.
    It uses tags to organize headings, paragraphs, images, links, tables, and forms.
  </p>

  <p>
    JavaScript is a programming language used to make web pages interactive and dynamic.
    It is used for buttons, animations, form validation, and other interactive features.
  </p>
</body>
</html>
```

6. **content1.html**

```
<html>
<head>
  <title>Contents</title>
</head>
<body>
  <h2>Web Technology</h2>
  <h3>HTML</h3>
  <p>
```

HTML (HyperText Markup Language) is used to create and structure web pages.

It uses tags to add headings, paragraphs, images, links, tables, and forms.

<h3>CSS</h3>

CSS (Cascading Style Sheets) is used to design and style web pages.

It controls colors, fonts, backgrounds, spacing, borders, and the layout of a webpage.

<h3>JavaScript</h3>

JavaScript is a programming language used to make web pages interactive and dynamic.

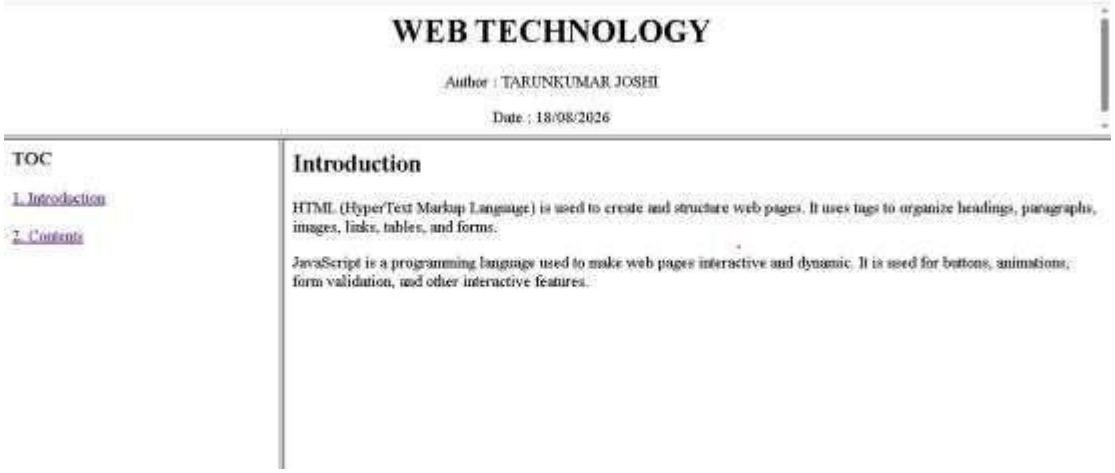
It is used for buttons, animations, form validation, pop-up messages, and updating webpage content.

</body>

</html>

OUTPUT:


WEB TECHNOLOGY
Author : TARUNKUMAR JOSHI
Date : 18/08/2026
TOC
1. Introduction
2. Contents
CONTENT


WEB TECHNOLOGY
Author : TARUNKUMAR JOSHI
Date : 18/08/2026
TOC
1. Introduction
2. Contents
Introduction
HTML (HyperText Markup Language) is used to create and structure web pages. It uses tags to organize headings, paragraphs, images, links, tables, and forms.
JavaScript is a programming language used to make web pages interactive and dynamic. It is used for buttons, animations, form validation, and other interactive features.


WEB TECHNOLOGY
Author : TARUNKUMAR JOSHI
Date : 18/08/2026
TOC
1. Introduction
2. Contents
Web Technology
HTML
HTML (HyperText Markup Language) is used to create and structure web pages. It uses tags to add headings, paragraphs, images, links, tables, and forms.
CSS
CSS (Cascading Style Sheets) is used to design and style web pages. It controls colors, fonts, backgrounds, spacing, borders, and the layout of a webpage.
JavaScript
JavaScript is a programming language used to make web pages interactive and dynamic. It is used for buttons, animations, form validation, pop-up messages, and updating webpage content.

Practical: 10

AIM: Create a simple form that will show all the INPUT METHODS available in HTML

CODE:

```
<html>

<head>

<title>Student Registration Form</title>

</head>

<body  bgcolor="yellow">

<center>

<h2>Student Registration Form</h2>

<form>

<table border="0" cellpadding="8">

<tr>

<td>Name</td>

<td><input type="text" name="name"></td>

</tr>

<tr>

<td>Password</td>

<td><input type="password" name="password"></td>

</tr>

<tr>

<td>Email</td>

<td><input type="email" name="email"></td>
```

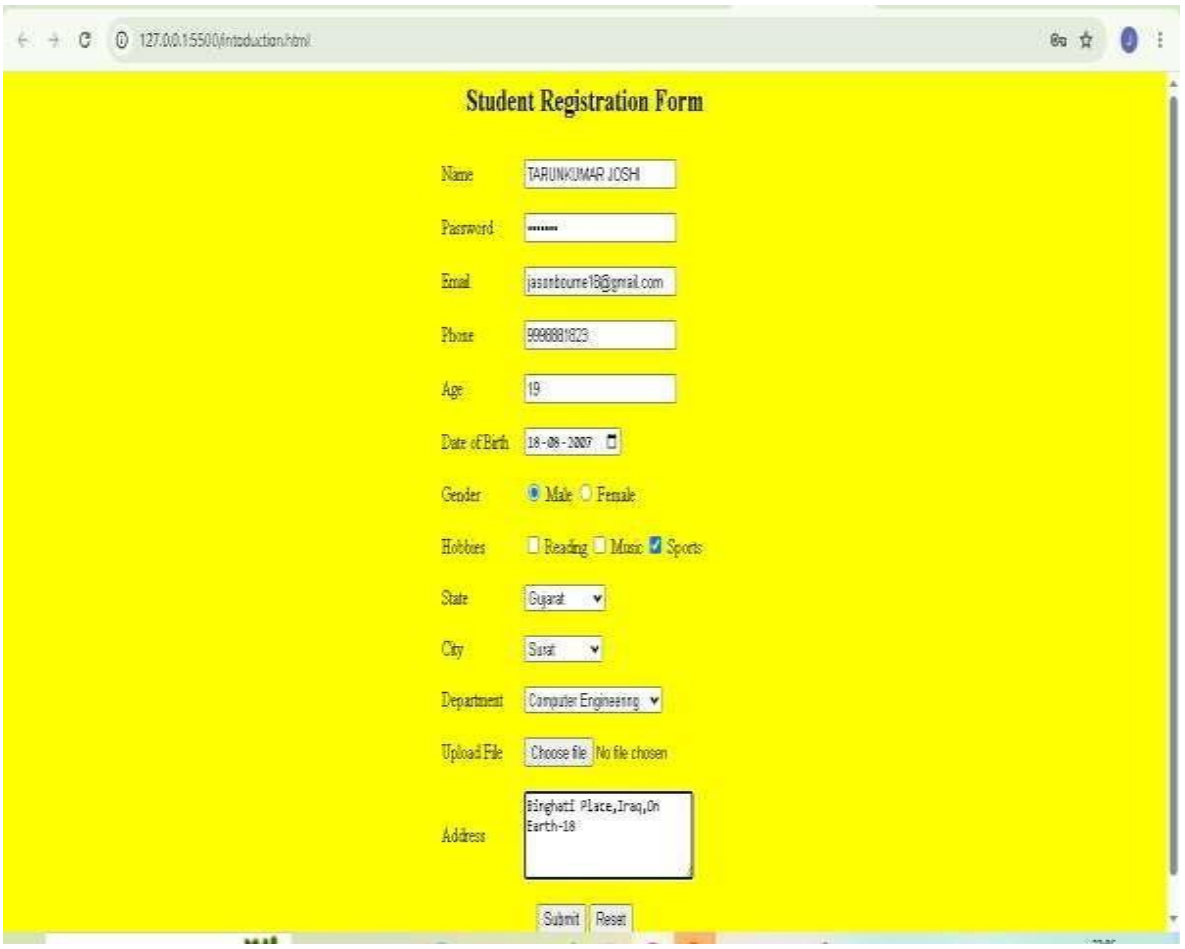
```
</tr>
 <tr>
<td>Phone</td>
<td><input type="tel" name="phone"></td>
</tr>
<tr>
<td>Age</td>
<td><input type="number"  name="age"></td>
</tr>
<tr>
<td>Date of Birth</td>
<td><input type="date" name="dob"></td>
</tr>
<tr>
<td>Gender</td>
<td>
<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female
</td>
</tr><tr><td>Hobbies</td>
<td>
<input type="checkbox" name="hobby"> Reading
<input type="checkbox" name="hobby"> Music
```

```
<input type="checkbox" name="hobby"> Sports
</td>
</tr>
<tr>
<td>State</td>
<td>
<select>
<option>Select State</option>
<option>Gujarat</option>
<option>Maharashtra</option>
<option>Rajasthan</option>
<option>Delhi</option>
 </select>
</td>
</tr>
<tr>
<td>City</td>
<td>
<select>
<option>Select City</option>
 <option>Surat</option>
<option>Ahmedabad</option>
<option>Vadodara</option>
```

```
<option>Rajkot</option>
</select>
</td>
</tr>
<tr>
<td>Department</td>
<td>
<select>
<option>Computer Engineering</option>
<option>Electrical Engineering</option>
<option>Mechanical Engineering</option>
 </select>
</td>
</tr>
<tr>
<td>Upload File</td>
<td><input type="file" name="file"></td>
</tr>
<tr>
<td>Address</td>
<td>
<textarea rows="4" cols="25"></textarea>
</td>
```

```
</tr><tr><td colspan="2" align="center">
<input type="submit" value="Submit">
<input type="reset" value="Reset">
</td>
</tr>
</table>
</form>
</center>
</body>
</html>
```

OUTPUT:


127.0.0.1:5500/introduction.html
Student Registration Form
Name
TARUNKUMAR JOSHI
Password
Email
jasonbourne18@gmail.com
Phone
9998881823
Age
19
Date of Birth
18-08-2007
Gender
Male
Female
Hobbies
Reading
Music
Sports
State
Gujarat
City
Surat
Department
Computer Engineering
Upload File
Choose file
No file chosen
Address
Birghati Place,Iraq,On Earth-18
Submit
Reset

Practical 1

AIM: Create simple application that will do following Declare And assign variable, Operators and expression in JavaScript, Looping in JavaScript, Declare an Array, User defined functions in Javascript, Built in functions in Javascript, Dialog boxes.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>JavaScript Basics</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f2f2f2;
      text-align: center;
      padding: 30px;
    }

    .container {
      background: white;
      max-width: 700px;
      margin: auto;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 0 10px gray;
    }

    button {
      padding: 10px 18px;
      margin: 8px;
      cursor: pointer;
```

```
            border: none;
            border-radius: 5px;
            background-color: #007bff;
            color: white;
        }

        button:hover {
            background-color: #0056b3;
        }

        #output {
            margin-top: 20px;
            padding: 15px;
            background-color: #eee;
            text-align: left;
            border-radius: 5px;
        }
    </style>
</head>

<body>

<div class="container">

    <h1>JavaScript Basics Demo</h1>

    <button onclick="showVariables()">Variables</button>
    <button onclick="showOperators()">Operators & Expression</button>
    <button onclick="showLoop()">Looping</button>
    <button onclick="showArray()">Array</button>
    <button onclick="showFunction()">User Defined Function</button>
```

```
        <button onclick="showBuiltIn()">Built-in Functions</button>
        <button onclick="showDialog()">Dialog Boxes</button>

        <div id="output">
            Click a button to see the result.
        </div>

    </div>

    <script>

function showVariables() {

        let name = "Rahul";
        let age = 20;
        const country = "India";
        document.getElementById("output").innerHTML =
            "<h3>Variables</h3>" +
            "Name: " + name + "<br>" +
            "Age: " + age + "<br>" +
            "Country: " + country;
    }

function showOperators() {

        let a = 10;
        let b = 5;
        let addition = a + b;
        let subtraction = a - b;
        let multiplication = a * b;
        let division = a / b;
        let remainder = a % b;
        document.getElementById("output").innerHTML =
```

```
            "<h3>Operators and Expressions</h3>" +
            "a = " + a + ", b = " + b + "<br><br>" +
            "Addition (a + b) = " + addition + "<br>" +
            "Subtraction (a - b) = " + subtraction + "<br>" +
            "Multiplication (a * b) = " + multiplication + "<br>" +
            "Division (a / b) = " + division + "<br>" +
            "Remainder (a % b) = " + remainder;
    }

function showLoop() {

        let result = "<h3>For Loop</h3>";
        for (let i = 1; i <= 5; i++) {
            result += "Number: " + i + "<br>";
        }
        document.getElementById("output").innerHTML = result;
    }

function showArray() {

        let fruits = ["Apple", "Banana", "Mango", "Orange"];
        let result = "<h3>Array</h3>";
        result += "Fruits: " + fruits.join(", ") + "<br><br>";
        for (let i = 0; i < fruits.length; i++) {
            result += "Fruit " + (i + 1) + ": " + fruits[i] + "<br>";
        }
        document.getElementById("output").innerHTML = result;
    }

function addNumbers(x, y) {
        return x + y;
    }
```

```
function showFunction() {

        let num1 = 20;
        let num2 = 30;
        let result = addNumbers(num1, num2);
        document.getElementById("output").innerHTML =
            "<h3>User Defined Function</h3>" +
            "First Number: " + num1 + "<br>" +
            "Second Number: " + num2 + "<br>" +
            "Result: " + result;
    }

    function showBuiltIn() {

        let text = "javascript";
        let upperText = text.toUpperCase();
        let length = text.length;
        let roundedNumber = Math.round(10.6);
        document.getElementById("output").innerHTML =
            "<h3>Built-in Functions</h3>" +
            "Original Text: " + text + "<br>" +
            "toUpperCase(): " + upperText + "<br>" +
            "length: " + length + "<br>" +
            "Math.round(10.6): " + roundedNumber;
    }

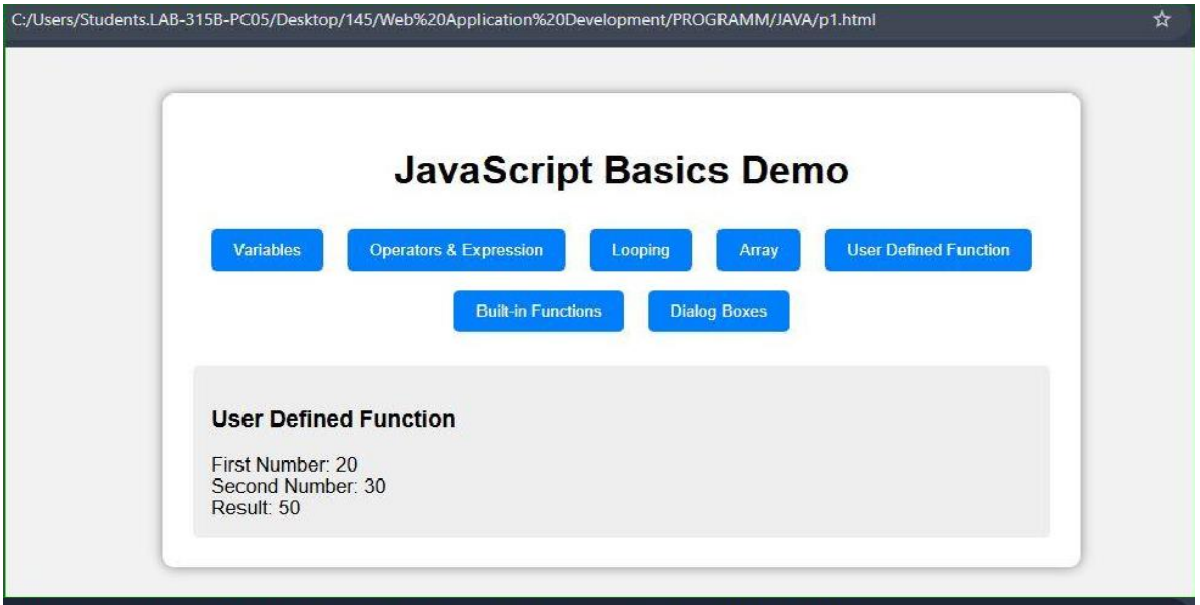

function showDialog() {

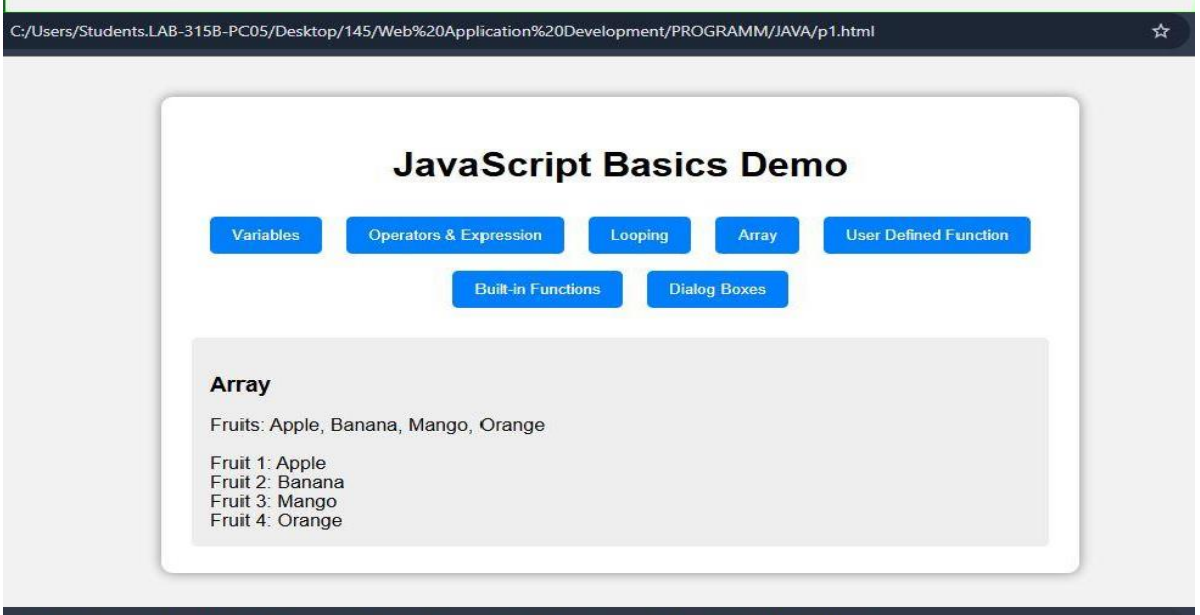
        alert("Welcome to the JavaScript Demo!");
        let name = prompt("Enter your name:");
```

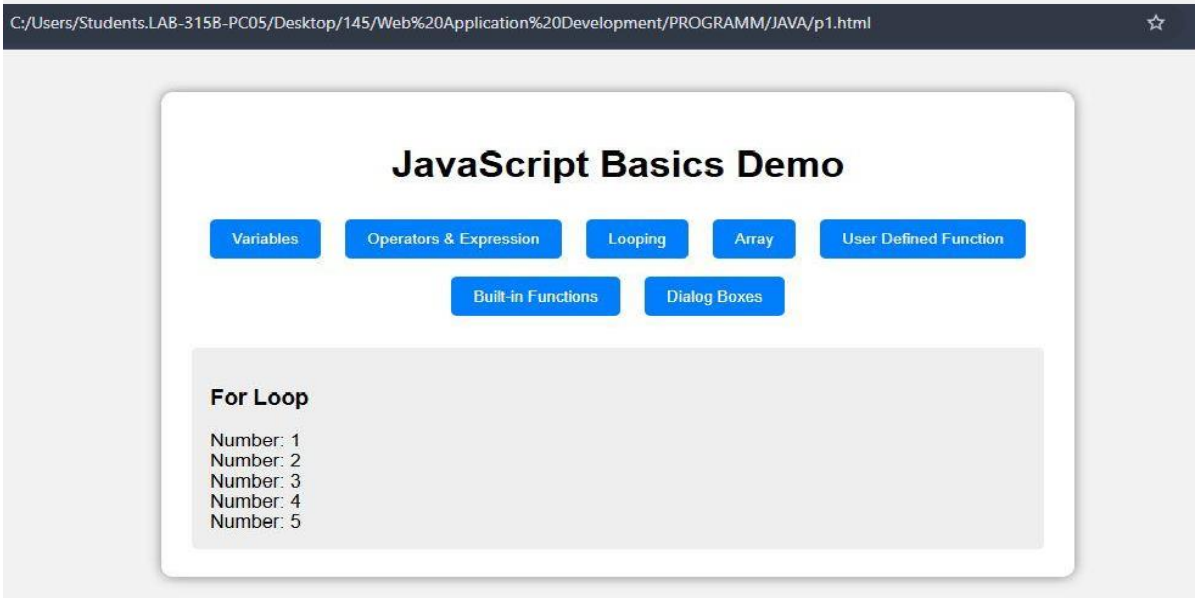
```
    let answer = confirm("Do you want to continue?");
    if (answer) {
        document.getElementById("output").innerHTML =
            "<h3>Dialog Boxes</h3>" +
            "Hello, " + name + "!<br>" +
            "You selected OK.";
    } else {
        document.getElementById("output").innerHTML =
            "<h3>Dialog Boxes</h3>" +
            "Hello, " + name + "!<br>" +
            "You selected Cancel.";
    }
}
</script>
</body>
</html>
```

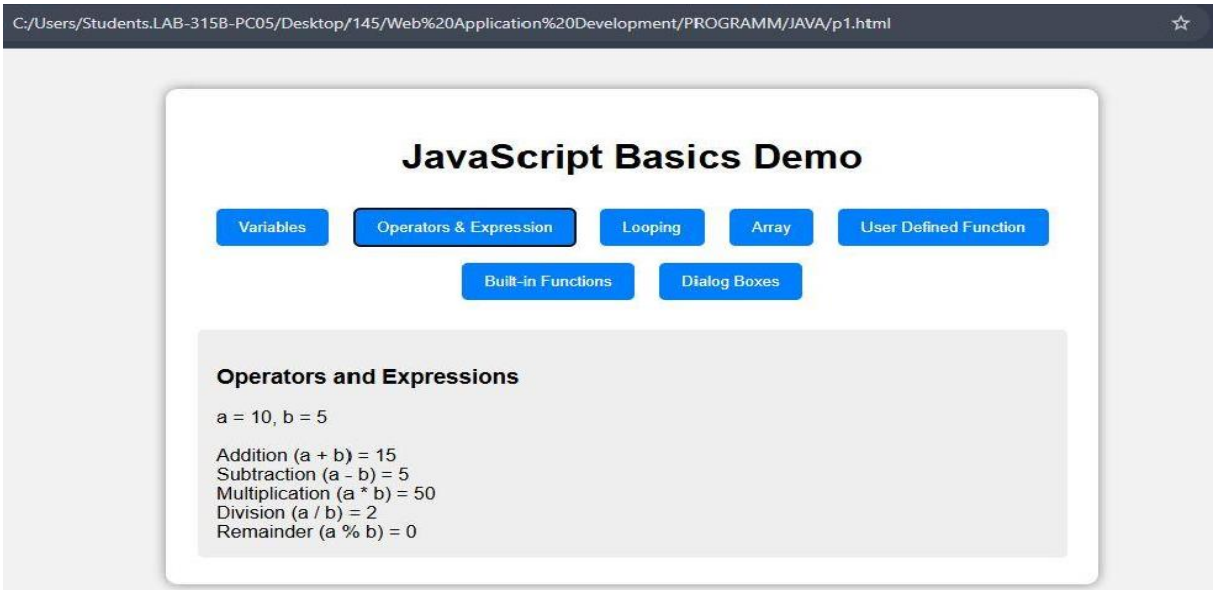
OUTPUT:

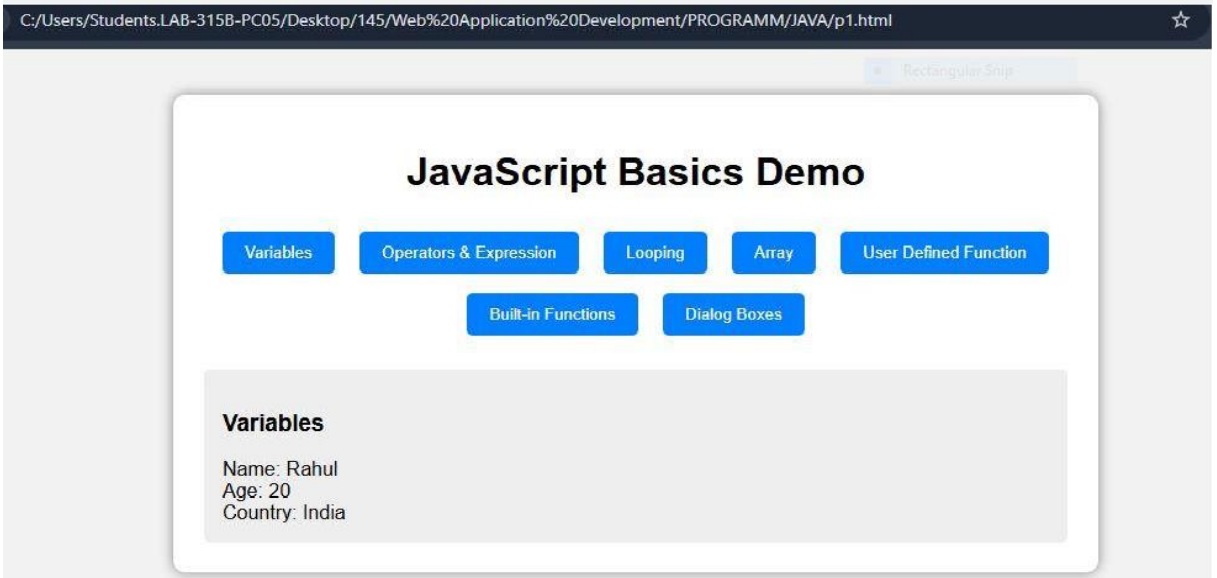


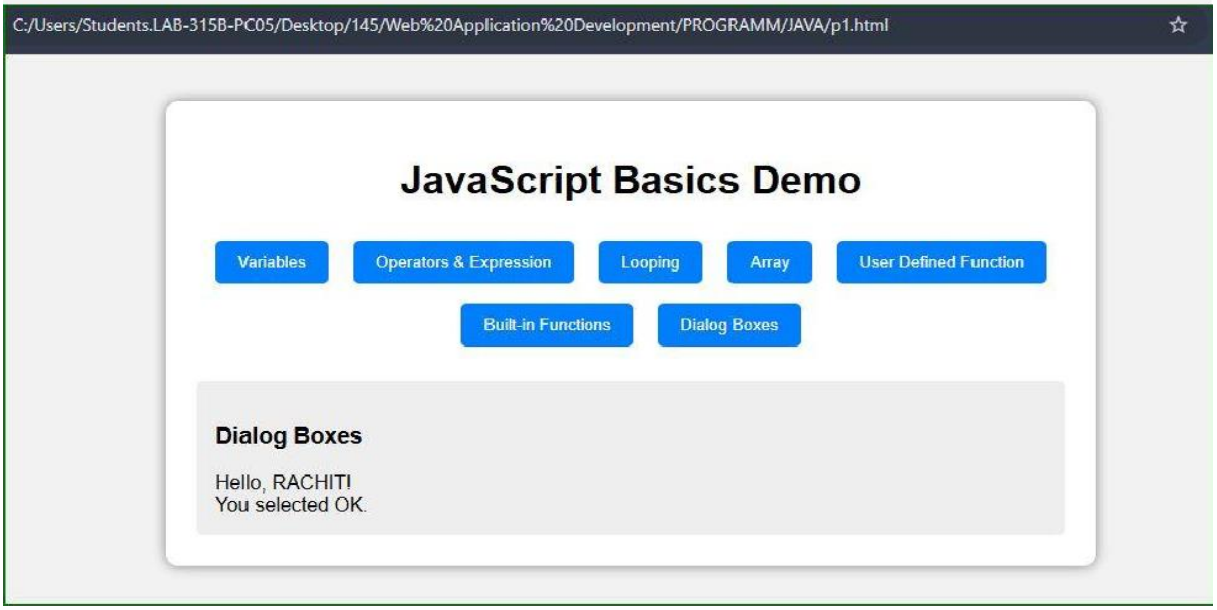

C:/Users/Students.LAB-315B-PC05/Desktop/145/Web%20Application%20Development/PROGRAMM/JAVA/p1.html
JavaScript Basics Demo
Variables
Operators & Expression
Looping
Array
User Defined Function
Built-in Functions
Dialog Boxes
User Defined Function
First Number: 20
Second Number: 30
Result: 50


C:/Users/Students.LAB-315B-PC05/Desktop/145/Web%20Application%20Development/PROGRAMM/JAVA/p1.html
JavaScript Basics Demo
Variables
Operators & Expression
Looping
Array
User Defined Function
Built-in Functions
Dialog Boxes
Array
Fruits: Apple, Banana, Mango, Orange
Fruit 1: Apple
Fruit 2: Banana
Fruit 3: Mango
Fruit 4: Orange


C:/Users/Students.LAB-315B-PC05/Desktop/145/Web%20Application%20Development/PROGRAMM/JAVA/p1.html
JavaScript Basics Demo
Variables
Operators & Expression
Looping
Array
User Defined Function
Built-in Functions
Dialog Boxes
For Loop
Number: 1
Number: 2
Number: 3
Number: 4
Number: 5


C:/Users/Students.LAB-315B-PC05/Desktop/145/Web%20Application%20Development/PROGRAMM/JAVA/p1.html
JavaScript Basics Demo
Variables
Operators & Expression
Looping
Array
User Defined Function
Built-in Functions
Dialog Boxes
Operators and Expressions
a = 10, b = 5
Addition (a + b) = 15
Subtraction (a - b) = 5
Multiplication (a * b) = 50
Division (a / b) = 2
Remainder (a % b) = 0


C:/Users/Students.LAB-315B-PC05/Desktop/145/Web%20Application%20Development/PROGRAMM/JAVA/p1.html
JavaScript Basics Demo
Variables
Operators & Expression
Looping
Array
User Defined Function
Built-in Functions
Dialog Boxes
Variables
Name: Rahul
Age: 20
Country: India


C:/Users/Students.LAB-315B-PC05/Desktop/145/Web%20Application%20Development/PROGRAMM/JAVA/p1.html
JavaScript Basics Demo
Variables
Operators & Expression
Looping
Array
User Defined Function
Built-in Functions
Dialog Boxes
Dialog Boxes
Hello, RACHIT!
You selected OK.

Practical 2

AIM: For the form created in HTML display the field data on the webpage.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Form Validation using JavaScript</title>

   <style>
      body {
         font-family: Arial, sans-serif;
         background-color: #f2f2f2;
      }

      .form-container {
         width: 400px;
         margin: 50px auto;
         background: white;
         padding: 25px;
         border-radius: 10px;
         box-shadow: 0 0 10px gray;
      }

      h2 {
         text-align: center;
      }

      label {
         display: block;
         margin-top: 12px;
      }

      input {
         width: 95%;
         padding: 8px;
         margin-top: 5px;
      }

      .gender {
         width: auto;
      }

      button {
         width: 100%;
         padding: 10px;
         margin-top: 20px;
         background-color: #007bff;
         color: white;
```

```
            border: none;
            border-radius: 5px;


cursor: pointer;
        }

        button:hover {
            background-color: #0056b3;
        }

        #message {
            margin-top: 15px;
            font-weight: bold;
        }
    </style>
</head>

<body>

<div class="form-container">

    <h2>Registration Form</h2>

    <form onsubmit="return validateForm()">

        <label>Name:</label>
        <input type="text" id="name">

        <label>Email:</label>
        <input type="text" id="email">

        <label>Password:</label>
        <input type="password" id="password">

        <label>Age:</label>
        <input type="number" id="age">

        <label>Mobile Number:</label>
        <input type="text" id="mobile">

        <label>Gender:</label>

        <input class="gender"
               type="radio"
               name="gender"
               value="Male">
        Male

        <input class="gender"
               type="radio"
```

```
            name="gender"
            value="Female">
        Female

       <br>

        <label>
           <input class="gender"
                type="checkbox"
                id="terms">
           I agree to the terms and conditions
        </label>

        <button type="submit">Submit</button>

    </form>

    <div id="message"></div>

</div>

<script>

function validateForm() {

    // Get values entered by user
    let name = document.getElementById("name").value.trim();
    let email = document.getElementById("email").value.trim();
    let password = document.getElementById("password").value;
    let age = document.getElementById("age").value;
    let mobile = document.getElementById("mobile").value.trim();

    let message = document.getElementById("message");

    if (name === "") {
        message.innerHTML = "Please enter your name.";
        message.style.color = "red";
        return false;
    }

    if (name.length < 3) {
        message.innerHTML = "Name must contain at least 3 characters.";
        message.style.color = "red";
        return false;
    }
    let emailPattern =
        /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if (email === "") {
        message.innerHTML = "Please enter your email.";
        message.style.color = "red";
```

```
    return false;
}

if (!emailPattern.test(email)) {

    message.innerHTML = "Please enter a valid email address.";
    message.style.color = "red";
    return false;
}

if (password === "") {
    message.innerHTML = "Please enter a password.";
    message.style.color = "red";
    return false;
}

if (password.length < 6) {
    message.innerHTML =
        "Password must contain at least 6 characters.";
    message.style.color = "red";
    return false;
}
if (age === "") {
    message.innerHTML = "Please enter your age.";
    message.style.color = "red";
    return false;
}

if (age < 18 || age > 100) {
    message.innerHTML =
        "Age must be between 18 and 100.";
    message.style.color = "red";
    return false;
}

let mobilePattern = /^[0-9]{10}$/;

if (!mobilePattern.test(mobile)) {
    message.innerHTML =
        "Mobile number must contain exactly 10 digits.";
    message.style.color = "red";
    return false;
}

let gender =
    document.querySelector(
        'input[name="gender"]:checked'
    );

if (gender === null) {
    message.innerHTML =
```

```
            "Please select your gender.";
        message.style.color = "red";
        return false;
    }

    let terms =
        document.getElementById("terms").checked;

    if (!terms) {
        message.innerHTML =
            "Please accept the terms and conditions.";
        message.style.color = "red";
        return false;
    }

    message.innerHTML =
        "Form submitted successfully!";

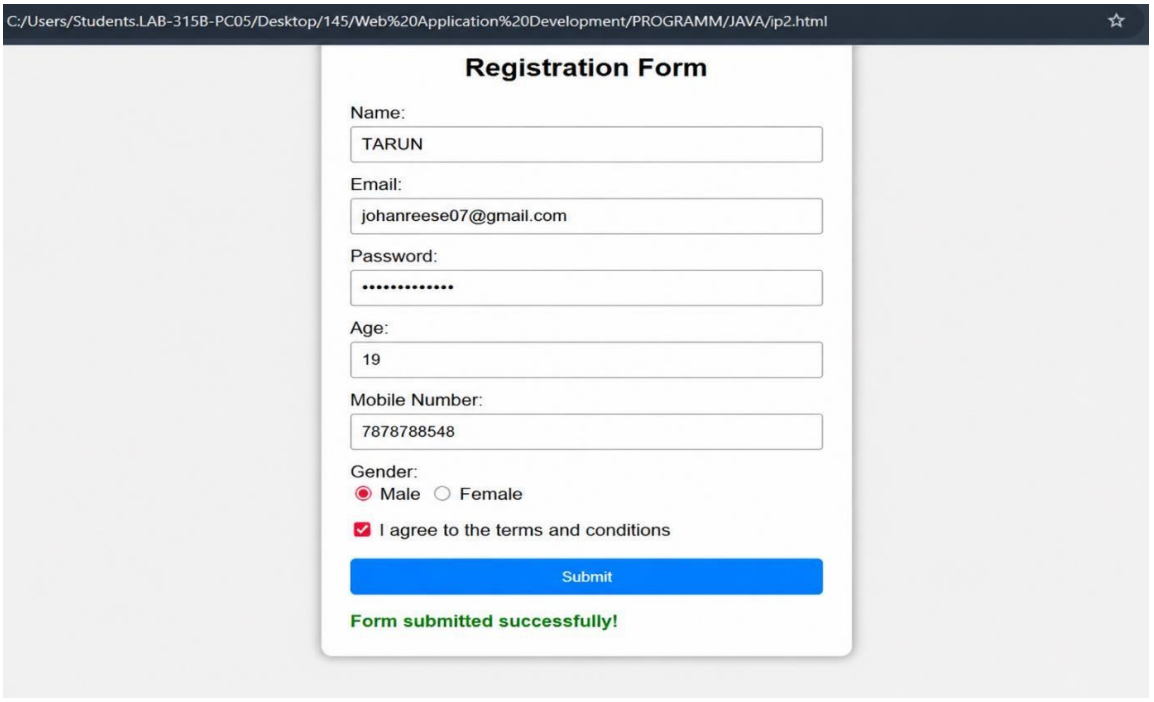
    message.style.color = "green";

    return false;
}

</script>

</body>
</html>
```

OUTPUT:



Form submitted successfully!

Entered Details

Name: TARUN
Email: johanreese@gmail.com
Password: tarun678@
Age: 18
Mobile Number: 9876543210
Gender: Male
Terms Accepted: Yes

Practical 3

AIM: A document contains two forms, named specifications and accessories. In the accessories form is a field named acc1(type=text). Write two different statements that set the contents of that field to "New value".

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Form Example</title>
</head>

<body>

<form name="specifications">
   <h3>Specifications Form</h3>
</form>

<form name="accessories">
   <h3>Accessories Form</h3>

   <label>Accessory 1:</label>
   <input type="text" name="acc1">

   <br><br>

   <button type="button" onclick="method1()">Method 1</button>
   <button type="button" onclick="method2()">Method 2</button>
</form>

<script>

function method1() {
   document.accessories.acc1.value = "New value";
}

function method2() {
   document.forms["accessories"].elements["acc1"].value = "New value";
}

</script>

</body>
</html>
```

OUTPUT:


127.0.0.1:5500/p3ja.html
Specifications Form
Accessories Form
Accessory 1: New value
Method 1
Method 2

Practical 4

AIM: Create a page that includes a select object to change the background color of the current page. The property that needs to be set is BGColor, Similar things for foreground color.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Change Page Colors</title>
</head>

<body>

   <h1>Change Page Colors</h1>

   <p>
      Select a background color and foreground color
      to change the appearance of this page.
   </p>

   <!-- Background Color -->
   <label>Background Color:</label>

   <select onchange="changeBackground(this.value)">
      <option value="white">White</option>
      <option value="lightblue">Light Blue</option>
      <option value="lightgreen">Light Green</option>
      <option value="yellow">Yellow</option>
      <option value="pink">Pink</option>
      <option value="black">Black</option>
   </select>
```

```
    <br><br>

    <!-- Foreground Color -->
    <label>Foreground Color:</label>

    <select onchange="changeForeground(this.value)">
        <option value="black">Black</option>
        <option value="white">White</option>
        <option value="red">Red</option>
        <option value="blue">Blue</option>
        <option value="green">Green</option>
        <option value="purple">Purple</option>
    </select>

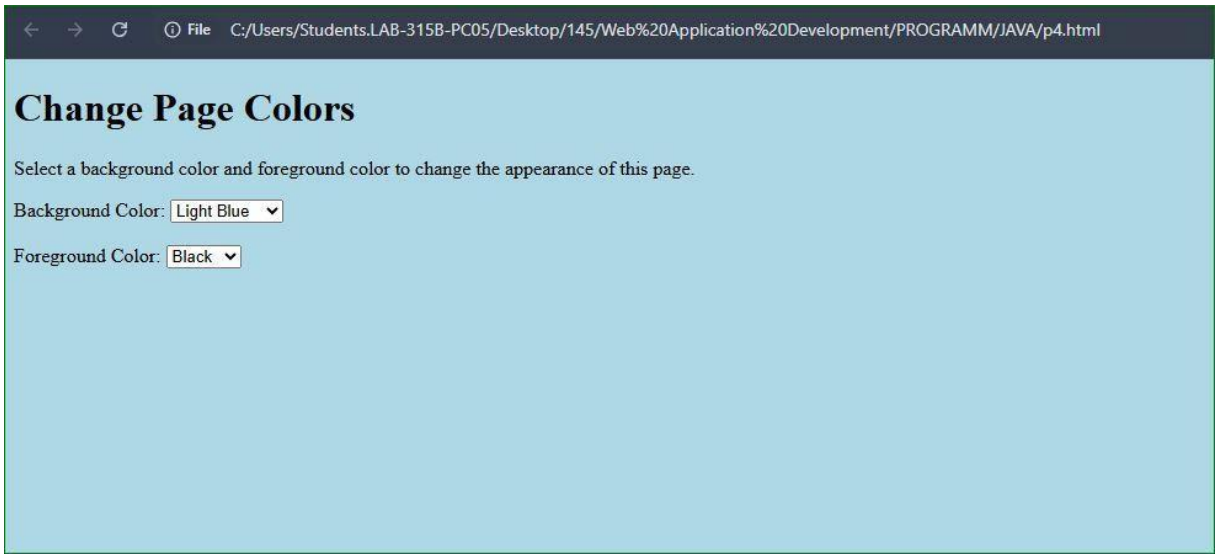
    <script>

        // Change background color
        function changeBackground(color) {
            document.bgColor = color;
        }

        // Change foreground/text color
        function changeForeground(color) {
            document.body.style.color = color;
        }

    </script>
</body>
</html>
```

OUTPUT:


File C:/Users/Students.LAB-315B-PC05/Desktop/145/Web%20Application%20Development/PROGRAMM/JAVA/p4.html
Change Page Colors
Select a background color and foreground color to change the appearance of this page.
Background Color: Light Blue
Foreground Color: Black

Practical 5

AIM: Write down simple JavaScript using timeout such that image will be changed after every 1 ms at a specified position.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>Image Movement using Timeout</title>

   <style>
      #myImage {
         position: absolute;
         width: 150px;
         height: 150px;
         top: 200px;
      }
   </style>
</head>

<body>

   <h2>Image Moving Left to Right and Right to Left</h2>

   <img id="myImage" src="zenitsurunleftright.jpg">

   <script>

      // Positions of image
      let positions = [
         "100px",     // Left
         "300px",
         "500px",
         "700px",
         "900px",
         "1100px"     // Right
      ];

      let index = 0;

      function moveLeftToRight() {

         let image = document.getElementById("myImage");

         // Set first image
         image.src = "zenitsurunleftright.jpg";

         // Change image position
         image.style.left = positions[index];
```

```
        index++;

        // Continue moving left to right
        if (index < positions.length) {

            setTimeout(moveLeftToRight, 500);

        } else {

            // Reached the right side
            // Start second image from right
            index = positions.length - 1;

            setTimeout(moveRightToLeft, 500);
        }
    }

    function moveRightToLeft() {

        let image = document.getElementById("myImage");

        // Set second image
        image.src = "zenitsurunrightleft.jpg";

        // Change image position
        image.style.left = positions[index];

        index--;

        // Continue moving right to left
        if (index >= 0) {

            setTimeout(moveRightToLeft, 500);

        } else {

            // Reached the left side
            // Start first image again
            index = 0;

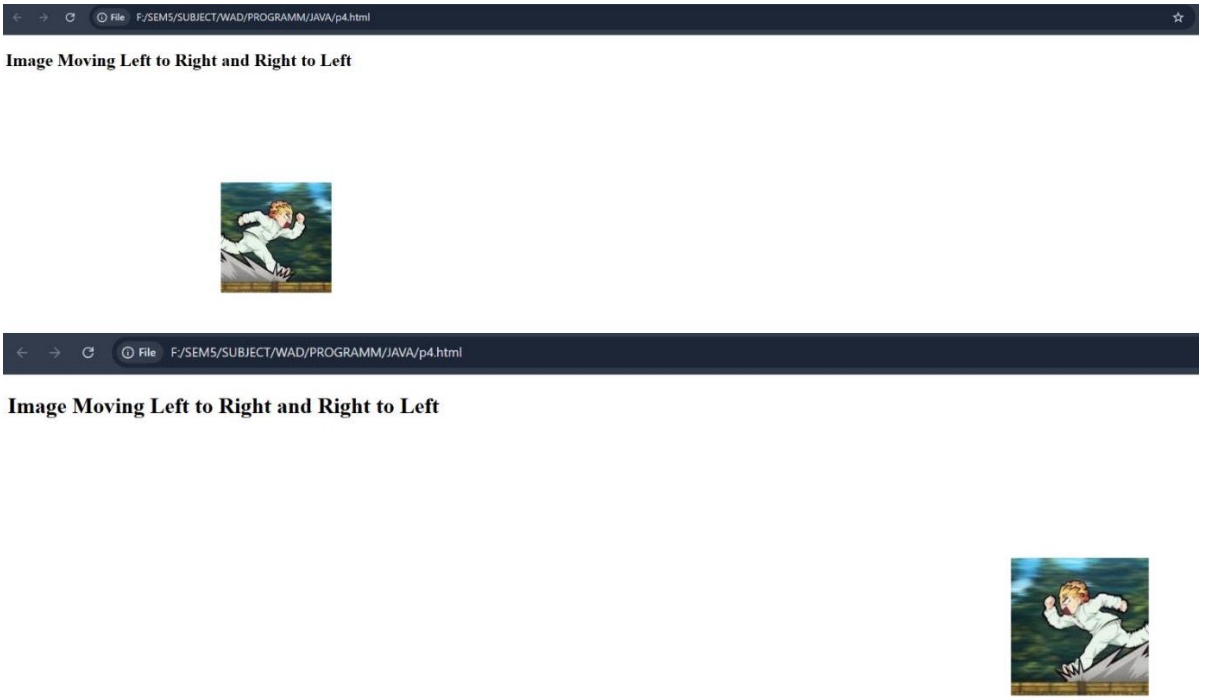
            setTimeout(moveLeftToRight, 500);
        }
    }

    moveLeftToRight();

  </script>

</body>
</html>
```

OUTPUT:


File F:/SEM5/SUBJECT/WAD/PROGRAMM/JAVA/p4.html
Image Moving Left to Right and Right to Left
File F:/SEM5/SUBJECT/WAD/PROGRAMM/JAVA/p4.html
Image Moving Left to Right and Right to Left

Practical 1

AIM: Demonstrate following attributes using CSS Color and background Font Text, Border, Margin and list.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>CSS Attributes</title>

  <style>
    /* Color */
    h1 {
      color: blue;
    }

    /* Background */
    body {
      background-color: lightyellow;
    }

    /* Font */
    p {
      font-family: Arial;
      font-size: 18px;
      font-weight: bold;
    }

    /* Text */
    .text {
      text-align: center;
```

```
            text-decoration: underline;
            text-transform: uppercase;
        }

        /* Border */
        .box {
            border: 2px solid black;
            padding: 10px;
        }

        /* Margin */
        .margin {
            margin: 30px;
        }

        /* List */
        ul {
            list-style-type: square;
        }
    </style>
</head>

<body>

    <h1>CSS Attributes Demonstration</h1>

    <p class="text">This is an example of text properties.</p>

    <div class="box margin">
        This box demonstrates Border and Margin.
```

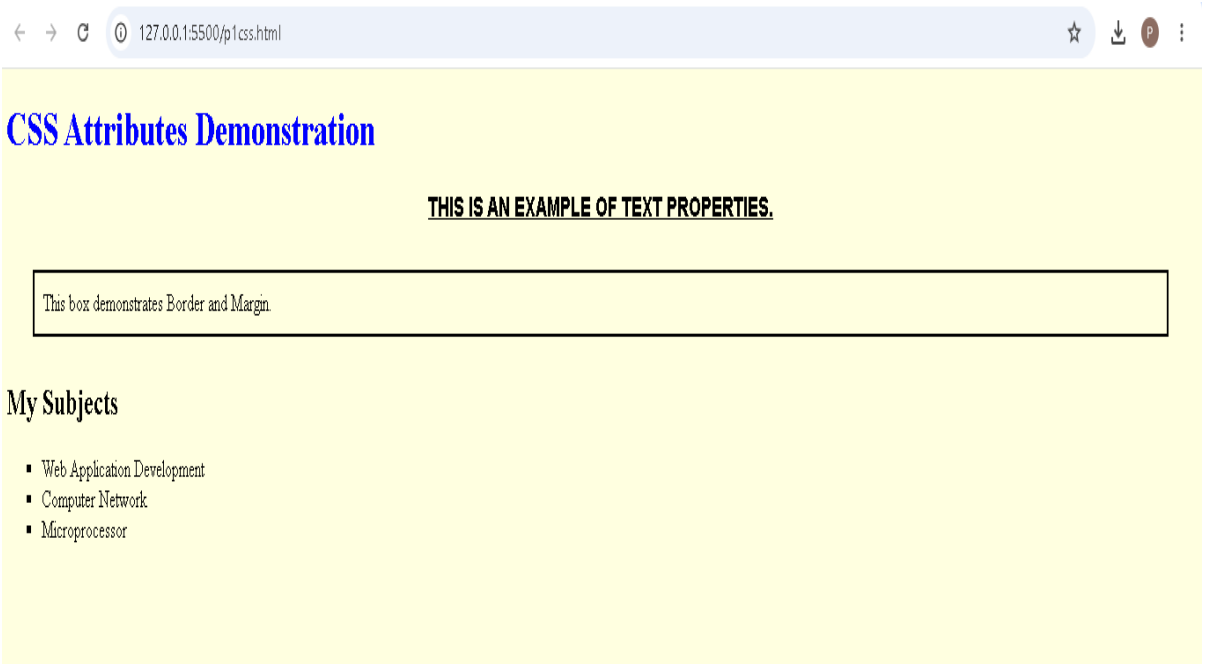
```
    </div>

    <h2>My Subjects</h2>

    <ul>
        <li>Web Application Development</li>
        <li>Computer Network</li>
        <li>Microprocessor</li>
    </ul>

</body>
</html>
```

OUTPUT:



Practical 2

AIM: Demonstrate use of external style sheet.

CODE:

```
<!DOCTYPE html>
<html>
<head>
   <title>External Style Sheet</title>

   <style>
      body {
         background-color: lightyellow;
         font-family: Arial;
         text-align: center;
      }

      h1 {
         color: blue;
         border: 2px solid black;
         padding: 10px;
      }

      p {
         color: green;
         font-size: 20px;
      }

      button {
         background-color: blue;
         color: white;
         padding: 10px 20px;
         border: none;
      }
   </style>
</head>

<body>

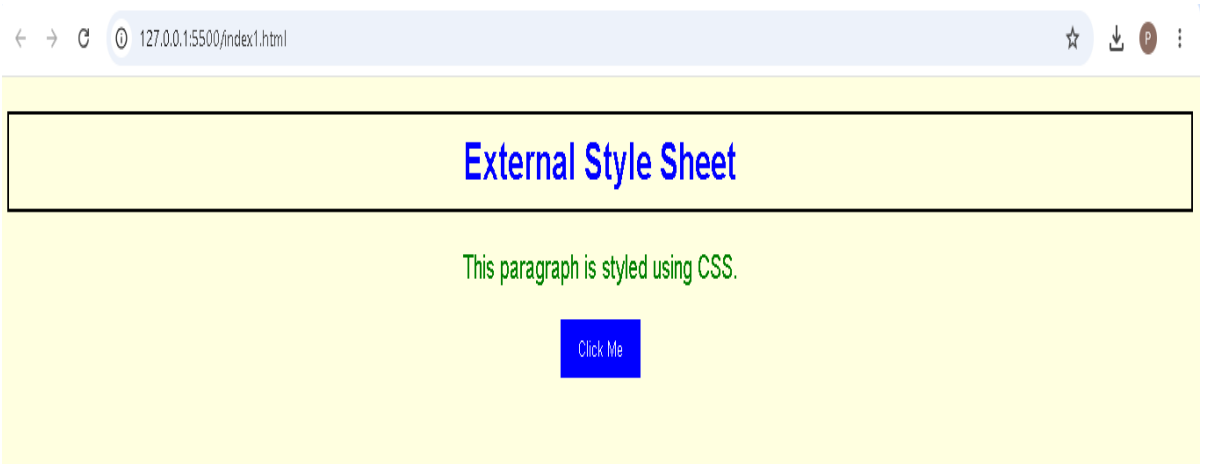
   <h1>External Style Sheet</h1>

   <p>This paragraph is styled using CSS.</p>

   <button>Click Me</button>

</body>
</html>
```

OUTPUT:


127.0.0.1:5500/index1.html
External Style Sheet
This paragraph is styled using CSS.
Click Me

Practical 2

AIM: Demonstrate use of external style sheet.

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>TARUN - Portfolio</title>

  <style>

    /* Body */
    body {
      background-color: lightyellow;
      font-family: Arial, sans-serif;
      margin: 0;
      color: #222;
    }

    /* Header */
    header {
      background-color: blue;
      color: white;
      text-align: center;
      padding: 30px;
    }

    header h1 {
      font-size: 40px;
      margin: 0;
    }

    header p {
      font-size: 18px;
    }

    /* Navigation */
    nav {
      background-color: #222;
      text-align: center;
      padding: 15px;
    }

    nav a {
      color: white;
      text-decoration: none;
      margin: 20px;
      font-weight: bold;
```

```
}

nav a:hover {
    color: yellow;
}

/* Sections */
section {
    background-color: white;
    margin: 25px;
    padding: 25px;
    border: 2px solid blue;
    border-radius: 10px;
}

/* Headings */
h2 {
    color: blue;
    text-align: center;
    text-decoration: underline;
}

/* About */
#about p {
    text-align: center;
    font-size: 18px;
    line-height: 1.6;
}

/* Skills */
ul {
    list-style-type: square;
    line-height: 2;
}

/* Projects */
.project {
    background-color: #f0f8ff;
    border: 2px solid #87ceeb;
    margin: 15px;
    padding: 15px;
    border-radius: 8px;
}

.project h3 {
    color: blue;
}

/* Contact */
#contact {
    text-align: center;
```

```
        }

        #contact p {
            font-size: 18px;
        }

        /* Footer */
        footer {
            background-color: #222;
            color: white;
            text-align: center;
            padding: 20px;
        }

    </style>
</head>

<body>

    <!-- Header -->
    <header>

        <h1>TARUN</h1>

        <p>Student | Web Developer | Programmer</p>

    </header>


    <!-- Navigation -->
    <nav>

        <a href="#about">About</a>
        <a href="#skills">Skills</a>
        <a href="#projects">Projects</a>
        <a href="#contact">Contact</a>

    </nav>


    <!-- About Section -->
    <section id="about">

        <h2>About Me</h2>

        <p>
            Hello! My name is TARUN.
            I am a student interested in web development
            and programming. I enjoy creating websites
            and learning new technologies.
        </p>
```

```
</section>


<!-- Skills Section -->
<section id="skills">

    <h2>My Skills</h2>

    <ul>
        <li>HTML</li>
        <li>CSS</li>
        <li>JavaScript</li>
        <li>Python</li>
        <li>Java</li>
    </ul>

</section>


<!-- Projects Section -->
<section id="projects">

    <h2>My Projects</h2>


    <div class="project">

        <h3>Registration Form</h3>

        <p>
            A registration form created using
            HTML, CSS and JavaScript.
        </p>

    </div>


    <div class="project">

        <h3>Student Management System</h3>

        <p>
            A simple project for managing
            student information and records.
        </p>

    </div>


    <div class="project">
```

```
        <h3>Personal Portfolio</h3>

        <p>
          A personal portfolio website created
          using HTML and CSS.
        </p>

      </div>

    </section>


    <!-- Contact Section -->
    <section id="contact">

      <h2>Contact Me</h2>

      <p>Email: johanreese07@gmail.com</p>

      <p>Mobile: 9876543210</p>

      <p>Location: Gujarat, India</p>

    </section>

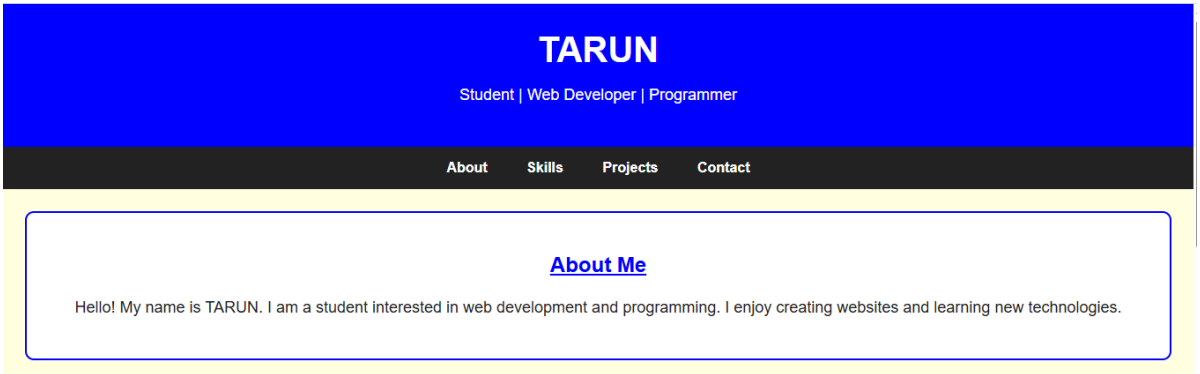

    <!-- Footer -->
    <footer>

      <p>© 2026 TARUN | My Portfolio</p>

    </footer>

</body>
</html>
```

OUTPUT:



My Skills

- HTML
- CSS
- JavaScript
- Python
- Java

My Projects

Registration Form

A registration form created using HTML, CSS and JavaScript.

View Project

Student Management System

A simple project for managing student information and records.

View Project

Personal Portfolio

A personal portfolio website created using HTML and CSS.

View Project

Contact Me

Email: johanreese07@gmail.com

Mobile: 9876543210

Location: Gujarat, India

Practical :- 1

Aim :- Write a Node.js program to create a basic HTTP server using the built-in http module.

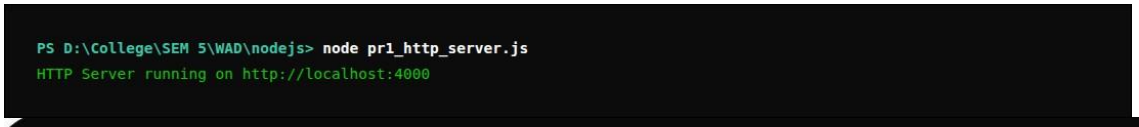
Code

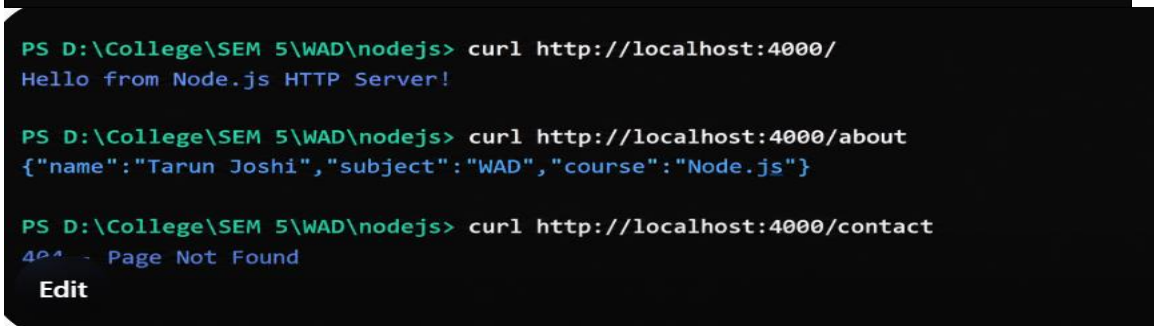
```
const http = require('http');

const server = http.createServer((req, res) => { if
  (req.url === '/' && req.method === 'GET') {
    res.writeHead(200, { 'Content-Type': 'text/plain' }); res.end('Hello
    from Node.js HTTP Server!');
  } else if (req.url === '/about' && req.method === 'GET') {
    res.writeHead(200, { 'Content-Type': 'application/json' });
    res.end(JSON.stringify({ name: 'Vaidik Gadhiya', subject: 'WAD', course: 'Node.js' }));
  } else {
    res.writeHead(404, { 'Content-Type': 'text/plain' }); res.end('404 -
    Page Not Found');
  }
});

const PORT = 4000; server.listen(PORT,
() => {
  console.log(`HTTP Server running on http://localhost:${PORT}`);
});
```

Output





Practical :- 2

Aim :- Create RESTful APIs using Node.js and Express.js supporting GET, POST, PUT, and DELETE HTTP methods.

Code

```
const express = require('express'); const
app = express();

app.use(express.json());

let students = [
  { id: 1, name: 'Vaidik Gadhiya', branch: 'Computer' },
  { id: 2, name: 'Krish Mehta', branch: 'IT' }
];

app.get('/api/students', (req, res) => {
  res.json(students);
});

app.get('/api/students/:id', (req, res) => { const
  id = parseInt(req.params.id);
  const student = students.find(s => s.id === id);
  if (!student) return res.status(404).json({ message: 'Student not found' });
  res.json(student);
});

app.post('/api/students', (req, res) => {
  const newStudent = {
    id: students.length > 0 ? Math.max(...students.map(s => s.id)) + 1 : 1, name:
    req.body.name,
    branch: req.body.branch
  };
  students.push(newStudent);
```

```
  res.status(201).json({ message: 'Student added successfully', student: newStudent });
});

app.put('/api/students/:id', (req, res) => { const
  id = parseInt(req.params.id);
  const student = students.find(s => s.id === id);
  if (!student) return res.status(404).json({ message: 'Student not found' });
  student.name = req.body.name || student.name;
  student.branch = req.body.branch || student.branch;
  res.json({ message: 'Student updated successfully', student });
});

app.delete('/api/students/:id', (req, res) => {
  const id = parseInt(req.params.id); students =
  students.filter(s => s.id !== id);
  res.json({ message: 'Student deleted successfully' });
});

const PORT = 5000; app.listen(PORT,
() => {
  console.log(`REST API Server running on http://localhost:${PORT}/api/students`);
});
```

Output

```
PS D:\College\SEM 5\WAD\nodejs> node pr2_rest_api.js
REST API Server running on http://localhost:5000/api/students
```


```
[
  {
    "id": 1,
    "name": "Tarun Joshi",
    "branch": "Computer"
  },
  {
    "id": 2,
    "name": "Krish Mehta",
    "branch": "IT"
  }
]
```


```
PS D:\College\SEM 5\WAD\nodejs> curl -X POST http://localhost:5000/api/students -H "Content-Type: application/json" -d
"{\"name\":\"Meet Joshi\",\"branch\":\"Mechanical\"}"
{"message":"Student added successfully","student":{"id":3,"name":"Meet Joshi","branch":"Mechanical"}}

PS D:\College\SEM 5\WAD\nodejs> curl -X PUT http://localhost:5000/api/students/1 -H "Content-Type: application/json" -
d "{\"branch\":\"Cyber Security\"}"
{"message":"Student updated successfully","student":{"id":1,"name":"Vaidik Gadhiya","branch":"Cyber Security"}}

PS D:\College\SEM 5\WAD\nodejs> curl -X DELETE http://localhost:5000/api/students/2
{"message":"Student deleted successfully"}
```

Practical :- 3

Aim :- Integrate MySQL database with Express backend and perform full CRUD operations.

Code

```
const express = require('express'); const
mysql = require('mysql2'); const app =
express();

app.use(express.json());

// 1. Database Connection
const db = mysql.createConnection({
  host: 'localhost',
  user: 'root',
  password: '',  database:
  'StudentDB'
});

db.connect((err) => { if
  (err) {
    console.log('MySQL Connection Error:', err.message);
  } else {
    console.log('Connected to MySQL Database: StudentDB');
  }
});

// CREATE: Insert a new student
app.post('/students', (req, res) => {
  const { name, email, rollno } = req.body;
  const sql = 'INSERT INTO student (name, email, rollno) VALUES (?, ?, ?)'; db.query(sql,
  [name, email, rollno], (err, result) => {
    if (err) return res.status(500).json(err);
```

```
    res.status(201).json({ message: 'Student created', id: result.insertId });
  });
});

// READ: Fetch all students
app.get('/students', (req, res) => {
  const sql = 'SELECT * FROM student';
  db.query(sql, (err, rows) => {
    if (err) return res.status(500).json(err);
    res.json(rows);
  });
});

// UPDATE: Update student info
app.put('/students/:id', (req, res) => {
const { name, email } = req.body;
  const sql = 'UPDATE student SET name = ?, email = ? WHERE id = ?';
  db.query(sql, [name, email, req.params.id], (err, result) => {
    if (err) return res.status(500).json(err);
    res.json({ message: 'Student updated successfully' });
  });
});

// DELETE: Delete student
app.delete('/students/:id', (req, res) => {
  const sql = 'DELETE FROM student WHERE id = ?';
  db.query(sql, [req.params.id], (err, result) => {
    if (err) return res.status(500).json(err);
    res.json({ message: 'Student deleted successfully' });
  });
});

const PORT = 3000; app.listen(PORT,
() => {
```

```
  console.log(`CRUD Server running on http://localhost:${PORT}/students`);
});
```

Output

```
PS D:\College\SEM 5\WAD\nodejs> node pr3_mysql_crud.js
Connected to MySQL Database: StudentDB
CRUD Server running on http://localhost:3000/students
```

```
PS D:\College\SEM 5\WAD\nodejs> curl -X POST http://localhost:3000/students -H "Content-Type: application/json" -d "
{"name":"Tarun Joshi","email":"tarun.joshi@gmail.com","rollno":"250765107008"}'
{"message":"Student created","id":1}

PS D:\College\SEM 5\WAD\nodejs> curl -X GET http://localhost:3000/students
[{"id":1,"name":"Tarun Joshi","email":"tarun.joshi@gmail.com","rollno":"250765107008"}]

PS D:\College\SEM 5\WAD\nodejs> curl -X PUT http://localhost:3000/students/1 -H "Content-Type: application/json" -d "
{"name":"Tarun Joshi","email":"tarun.joshi@gmail.com","rollno":"250765107008"}
{"message":"Student updated successfully"}

PS D:\College\SEM 5\WAD\nodejs> curl -X DELETE http://localhost:3000/students/1
{"message":"Student deleted successfully"}
```

Practical :- 1

Aim :- Develop a basic React.js application using Functional Components and JSX to display dynamic profile information**.**

Code:

```
import { useState } from 'react' function

ProfileCard() {

const student = {

name: "Tarun Joshi",

enrollment: "240760107186",

branch: "Computer Engineering",

college: "SSASIT",

email: "tarunjoshi@gmail.com"

};

return (

<section>

<h1>My Profile</h1>

<p>Name: {student.name}</p>

<p>Enrollment: {student.enrollment}</p>

<p>Branch: {student.branch}</p>

<p>College: {student.college}</p>

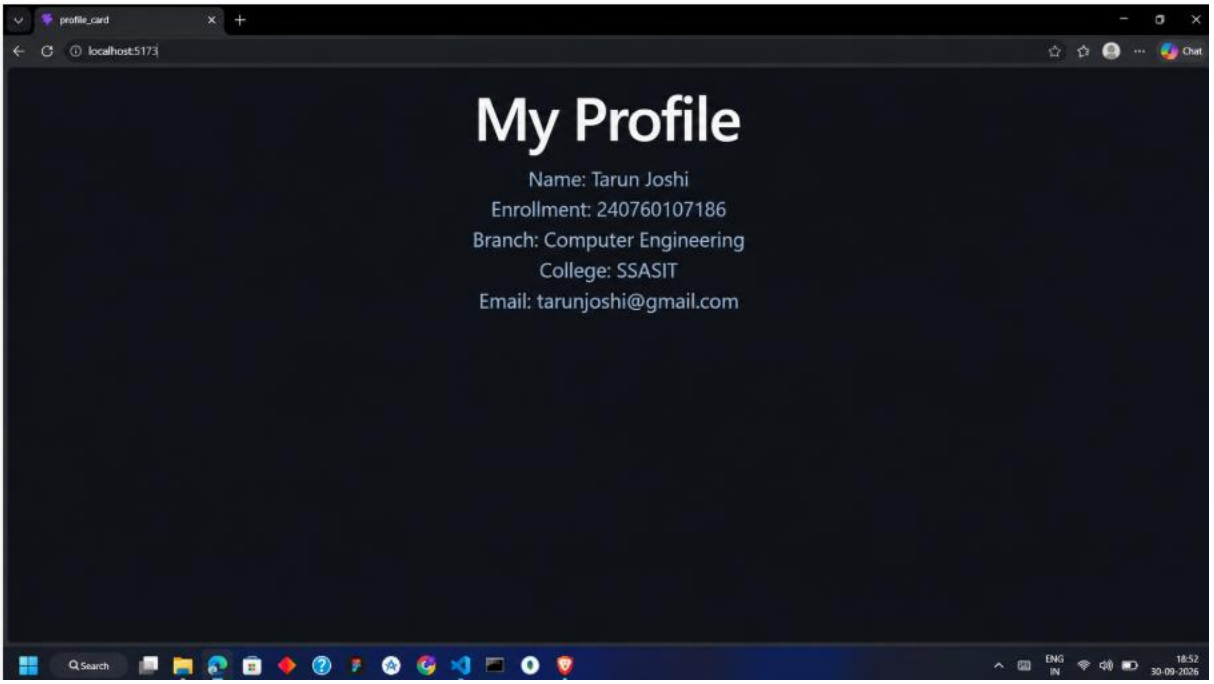
<p>Email: {student.email}</p>

</section>

);
}
function App() {

return <ProfileCard />;
}
export default App;
```

Output:


profile_card
localhost:5173
Chat
My Profile
Name: Tarun Joshi
Enrollment: 240760107186
Branch: Computer Engineering
College: SSASIT
Email: tarunjoshi@gmail.com
Search
ENG
IN
18:52
30-09-2026

Practical : 2

Aim : Create a dynamic Todo List application allowing users to add, complete, and delete tasks.

Code:

```
import { useState } from "react";

function App() {

const [task, setTask] = useState(""); const [tasks,

setTasks] = useState([]);

function addTask() {

if (task.trim()) {

setTasks([...tasks, { text: task, done: false }]);

setTask("");

}

}

function toggleTask(index) {


const updated = [...tasks];

updated[index].done = !updated[index].done;

setTasks(updated);

}

function removeTask(index)

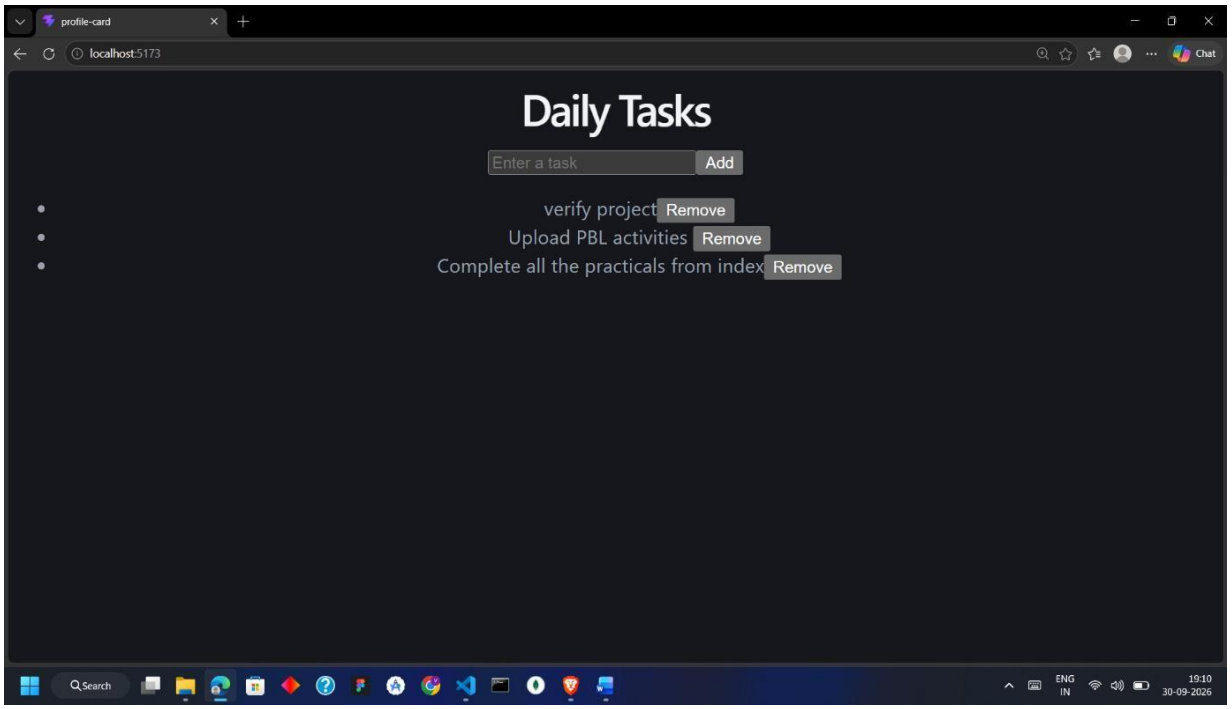

{

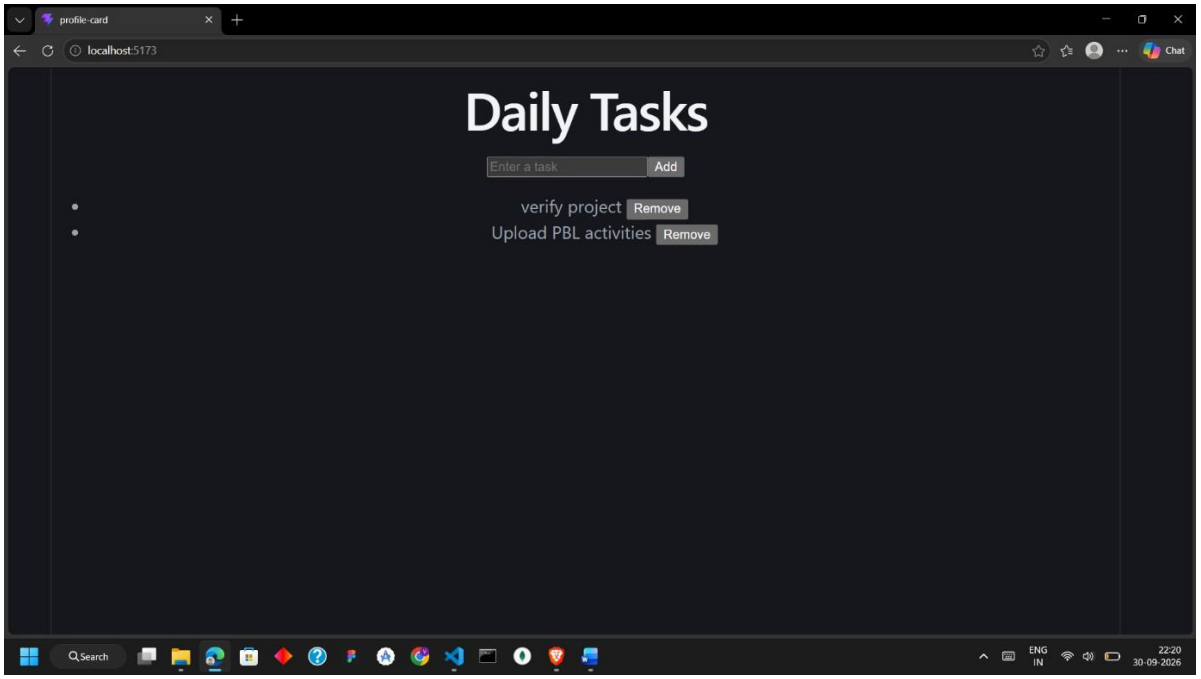

setTasks(tasks.filter((_, i) => i !== index));

}
```

```
return (
<div>
<h1>Daily Tasks</h1>
<input
value={task} placeholder="Enter a task"
onChange={(e) => setTask(e.target.value)}
/>
<button onClick={addTask}>Add</button>
<ul>
{tasks.map((item, index) => (
<li key={index}>
<span
onClick={() => toggleTask(index)}
style={{ textDecoration: item.done ? "line-through" : "none" }}
>
{item.text}
</span>
<button onClick={() => removeTask(index)}>Remove</button>
</li>
))}
</ul>
</div>
);
}
export default App;
```

Output :


profile-card
localhost:5173
Chat
Daily Tasks
Enter a task
Add
verify project
Remove
Upload PBL activities
Remove
Complete all the practicals from index
Remove
Search
ENG
IN
19:10
30-09-2026


profile-card
localhost:5173
Chat
Daily Tasks
Enter a task
Add
verify project
Remove
Upload PBL activities
Remove
Search
ENG
IN
22:20
30-09-2026

Practical : 3

Aim : Build a user registration form with controlled components, multi-input field binding, and real-time validation error messages.

Code:

```
import { useState } from "react";

function App() {

const [form, setForm] = useState({

name: "",

email: "",

password: "",

age: ""

});

const [errors, setErrors] = useState({}); function

handleChange(e)

 {

const { name, value } = e.target; setForm({ ...form,

[name]: value }); let error = "";

if (name === "name" && value.trim() === "")

 {

error = "Name is required";

}

else if (name === "email" && !value.includes("@"))

 {

error = "Enter a valid email";

} else if (name === "password" && value.length < 6)
```

```
{
error = "Password must have 6 characters";
} else if (name === "age" && Number(value) < 18)
{
 error = "Age must be 18 or above";
}
setErrors({ ...errors, [name]: error });
}
function submitForm(e) {
 e.preventDefault();
if (Object.values(errors).some(error => error)) return;
alert("Registration successful!");
}
function resetForm() {
setForm({ name: "", email: "", password: "", age: "" });

setErrors({});
}
return (
<div style={{ width: "400px", margin: "40px auto", fontFamily: "Arial" }}>
<h1>Student Registration</h1>
<form onSubmit={submitForm}>
<label>Name</label><br />
<input name="name" value={form.name} onChange={handleChange}
/><br />
<small>{errors.name}</small><br /><br />
```

```
<label>Email</label><br />

<input name="email" value={form.email} onChange={handleChange}
/><br />

<small>{errors.email}</small><br /><br />

<label>Password</label><br />

<input type="password" name="password" value={form.password}
onChange={handleChange} /><br />

<small>{errors.password}</small><br /><br />


<label>Age</label><br />

<input type="number" name="age" value={form.age}
onChange={handleChange} /><br />

<small>{errors.age}</small><br /><br />


<button type="submit">Register</button>

<button type="button" onClick={resetForm} style={{ marginLeft: "15px"
}}>Reset</button>

</form>

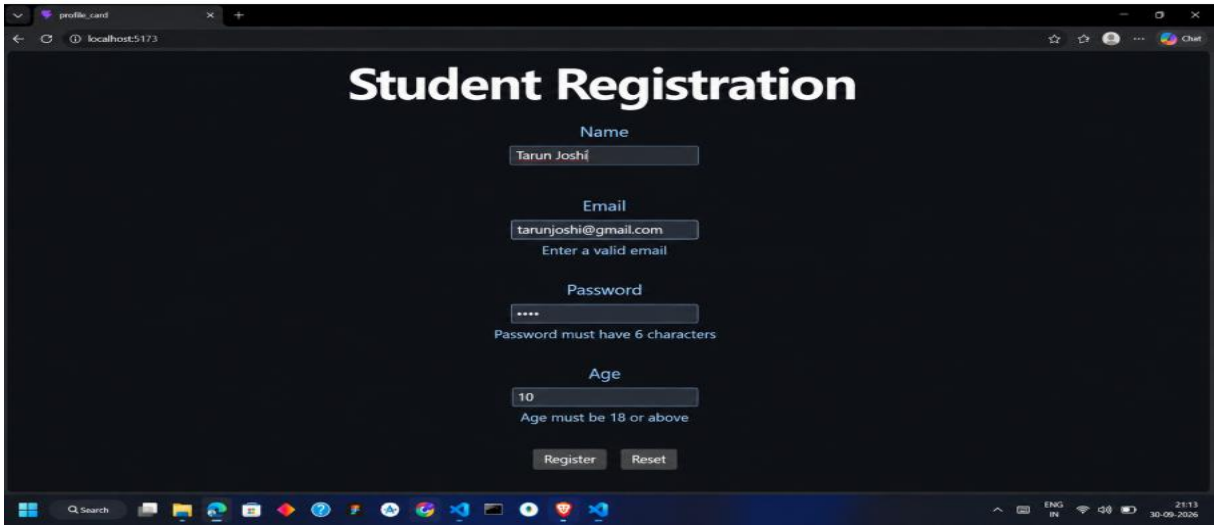
</div>

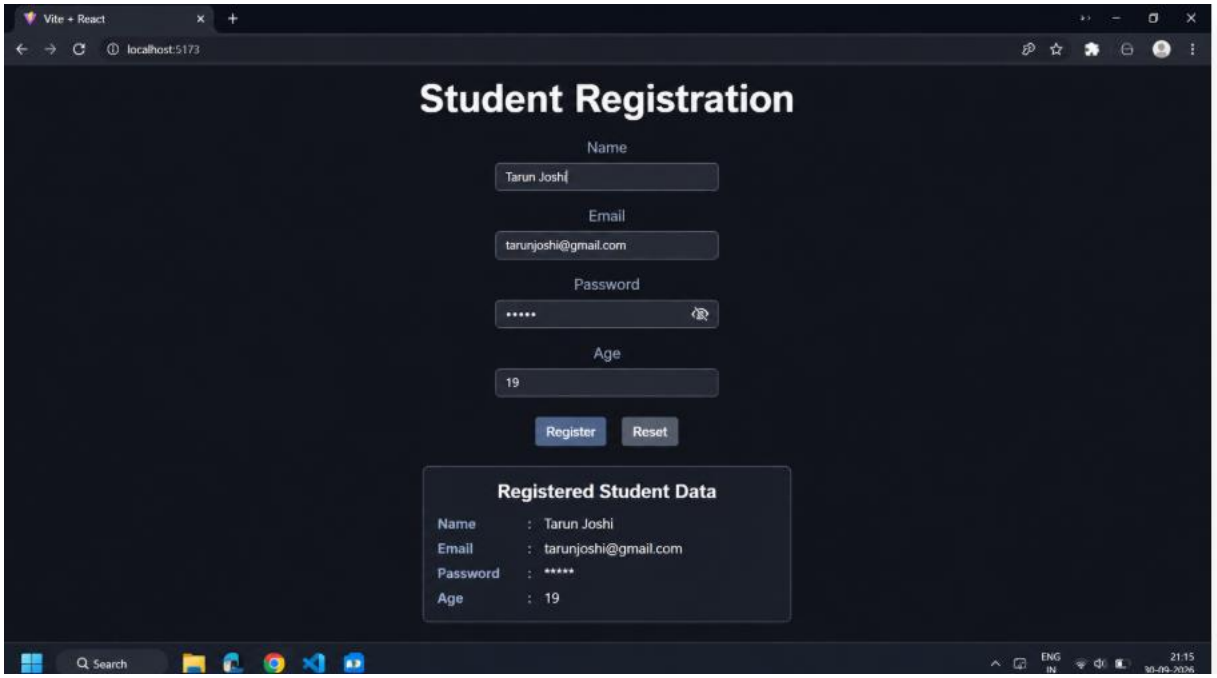
);

}

export default App
```

Output:


profile_card
localhost:5173
Student Registration
Name
Tarun Joshi
Email
tarunjoshi@gmail.com
Enter a valid email
Password
Password must have 6 characters
Age
10
Age must be 18 or above
Register
Reset


Vite + React
localhost:5173
Student Registration
Name
Tarun Joshi
Email
tarunjoshi@gmail.com
Password
Age
19
Register
Reset
Registered Student Data
Name : Tarun Joshi
Email : tarunjoshi@gmail.com
Password : *****
Age : 19

Practical : 4

Aim : Integrate frontend and backend services. Build a full Frontend-Backend integrated application that connects React.js UI with Node.js/Express REST APIs to fetch and mutate database records.

Code:

Backend

```
const express = require("express");
const mysql = require("mysql2");
const cors = require("cors");

const server = express();
server.use(cors());
server.use(express.json());

const database = mysql.createConnection({
    host: "localhost",
    port: 3306,
    user: "ashish",
    password: "ashish123",
    database: "ashish"
});

database.connect(err => {
    if (err) {
        console.log("Database connection error");
        return;
    }
    console.log("Database connected successfully");
});
```

```
server.get("/", (req, res) => {
    res.send("Student API is working");
});

server.get("/students", (req, res) => {
    database.query("SELECT * FROM students", (err, rows) => {
        if (err) return res.status(500).json({ message: "Unable to load students" });
        res.json(rows);
    });
});

server.post("/students", (req, res) => {
    const { name, email, age } = req.body;

    if (!name || !email || !age) {
        return res.status(400).json({ message: "Please fill all fields" });
    }

    database.query(
        "INSERT INTO students (name,email,age) VALUES (?,?,?)",
        [name, email, age],
        (err, result) => {
            if (err) return res.status(500).json({ message: "Unable to add student" });
            res.status(201).json({
                message: "Student added successfully",
                id: result.insertId
            });
        }
    );
});

server.put("/students/:id", (req, res) => {
    const { name, email, age } = req.body;
```

```
        database.query(
            "UPDATE students SET name=?, email=?, age=? WHERE id=?",
            [name, email, age, req.params.id],
            (err, result) => {
                if (err) return res.status(500).json({ message: "Unable to update student" });
                if (result.affectedRows === 0) {
                    return res.status(404).json({ message: "Student not found" });

 }
 res.json({ message: "Student updated successfully" });
}
);
});

    server.delete("/students/:id", (req, res) => {
        database.query(
            "DELETE FROM students WHERE id=?",
            [req.params.id],
            (err, result) => {
                if (err) return res.status(500).json({ message: "Unable to delete student" });
                if (result.affectedRows === 0) {
                    return res.status(404).json({ message: "Student not found" });
 }
 res.json({ message: "Student deleted successfully" }
 );
});
});
    server.listen(3000, () => {
        console.log("Server running at http://localhost:3000");
    });
```

Frontend

```
import { useEffect, useState } from "react";

function App() {
  const [records, setRecords] = useState([]);
  const [student, setStudent] = useState({
    name: "",
    email: "",
    age: ""
  });
  const [selectedId, setSelectedId] = useState(null);
  const [status, setStatus] = useState("");

  const fetchStudents = () => {
    fetch("http://localhost:3000/students")
      .then(res => res.json())
      .then(data => setRecords(data))
      .catch(() => setStatus("Server connection failed"));
  };

  useEffect(() => {
    fetchStudents();
  }, []);

  function changeInput(e) {
    setStudent({
      ...student,
      [e.target.name]: e.target.value
    });
  }

  function submitStudent(e) {
    e.preventDefault();
```

```
    const url = selectedId
      ? `http://localhost:3000/students/${selectedId}`
      : "http://localhost:3000/students";

    fetch(url, {
      method: selectedId ? "PUT" : "POST",
      headers: {
        "Content-Type": "application/json"
      },
      body: JSON.stringify(student)
    })
      .then(res => res.json())
      .then(data => {
        setStatus(data.message);
        clearForm();
        fetchStudents();
      });
  }

  function selectStudent(item) {
    setSelectedId(item.id);
    setStudent({
      name: item.name,
      email: item.email,
      age: item.age
    });
  }

  function removeStudent(id) {
    fetch(`http://localhost:3000/students/${id}`, {
      method: "DELETE"
    })
      .then(res => res.json())
      .then(data => {
```

```
            setStatus(data.message);
            fetchStudents();
        });
    }

    function clearForm() {
      setStudent({ name: "", email: "", age: "" });
      setSelectedId(null);
    }

    return (
      <div style={styles.page}>
        <h1>Student Records</h1>

        <div style={styles.box}>
          <h2>{selectedId ? "Edit Student" : "Add Student"}</h2>

          <form onSubmit={submitStudent}>
            <input
              name="name"
              placeholder="Student Name"
              value={student.name}
              onChange={changeInput}
              style={styles.input}
              required
            />
            <input
              name="email"
              type="email"
              placeholder="Email Address"
              value={student.email}
              onChange={changeInput}
              style={styles.input}
              required
```

```
        />
        <input
          name="age"
          type="number"
          placeholder="Age"
          value={student.age}
          onChange={changeInput}
          style={styles.input}
          required
        />

        <button type="submit" style={styles.mainButton}>
          {selectedId ? "Save Changes" : "Add Student"}
        </button>

        {selectedId && (
          <button
            type="button"
            onClick={clearForm}
            style={styles.cancelButton}
          >
            Cancel
          </button>
        )}
      </form>

      {status && <p style={styles.status}>{status}</p>}
    </div>

    <div style={styles.box}>
      <h2>Student List</h2>

      <table style={styles.table}>
        <thead>
```

```
        <tr>
          <th>ID</th>
          <th>Name</th>
          <th>Email</th>
          <th>Age</th>
          <th>Actions</th>
        </tr>
      </thead>

      <tbody>
        {records.map(item => (
          <tr key={item.id}>
            <td>{item.id}</td>
            <td>{item.name}</td>
            <td>{item.email}</td>
            <td>{item.age}</td>
            <td>
              <button
                onClick={() => selectStudent(item)}
                style={styles.editButton}
              >
                Edit
              </button>

              <button
                onClick={() => removeStudent(item.id)}
                style={styles.deleteButton}
              >
                Delete
              </button>
            </td>
          </tr>
        ))}
      </tbody>
```

```
            </table>

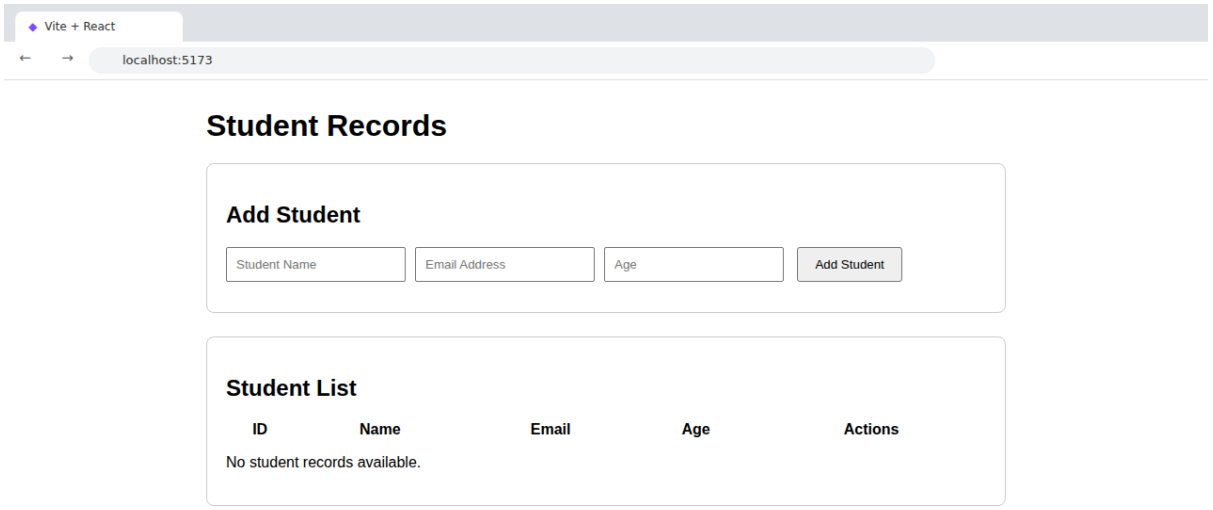
            {records.length === 0 && <p>No student records available.</p>}
          </div>
        </div>
    );
}

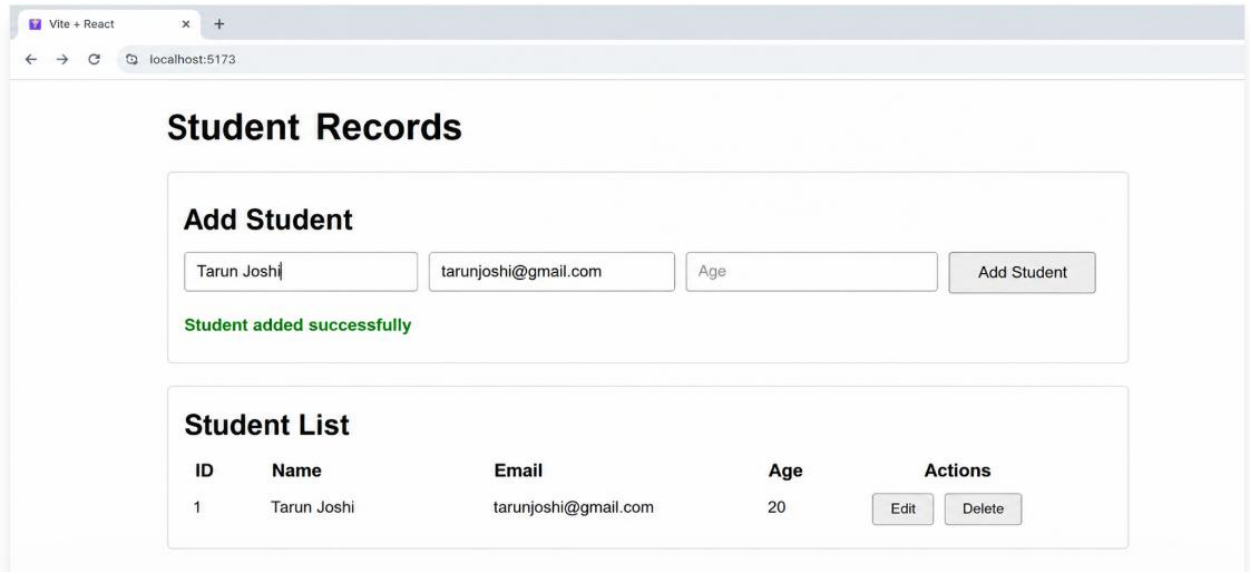
const styles = {
    page: {
        width: "850px",
        margin: "30px auto",
        fontFamily: "Arial"
    },
    box: {
        border: "1px solid #ccc",
        padding: "20px",
        marginBottom: "25px",
        borderRadius: "8px"
    },
    input: {
        padding: "10px",
        marginRight: "10px",
        marginBottom: "12px"
    },
    mainButton: {
        padding: "10px 18px",
        cursor: "pointer"
    },
    cancelButton: {
        padding: "10px 18px",
        marginLeft: "12px",
        cursor: "pointer"
    },
```

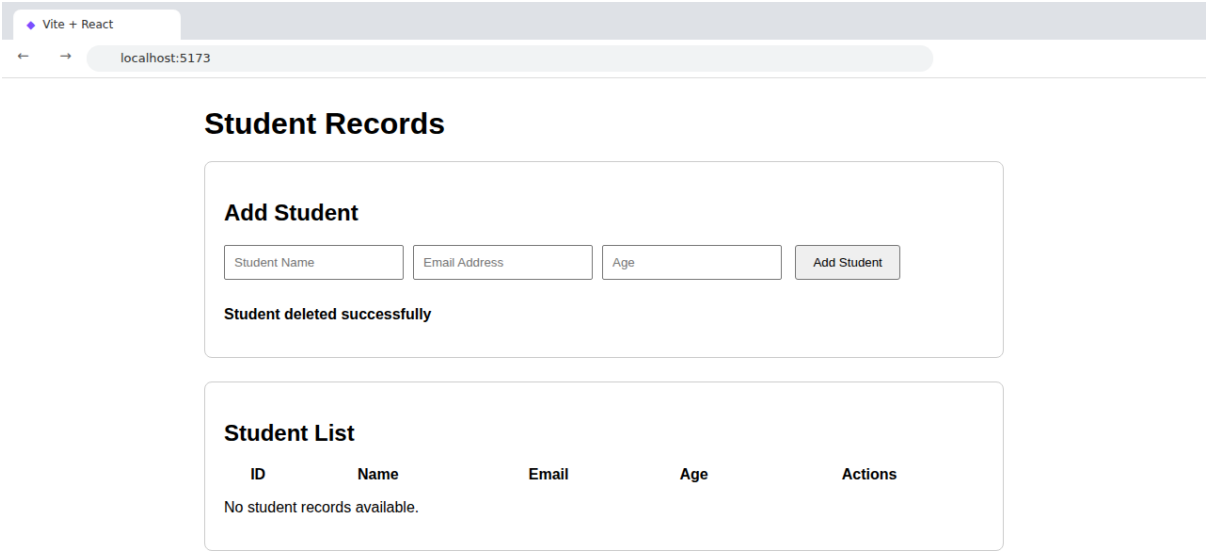
```
  editButton: {
    padding: "6px 12px",
    marginRight: "10px",
    cursor: "pointer"
  },
  deleteButton: {
    padding: "6px 12px",
    cursor: "pointer"
  },
  table: {
    width: "100%",
    borderCollapse: "collapse"
  },
  status: {
    fontWeight: "bold"
  }
};

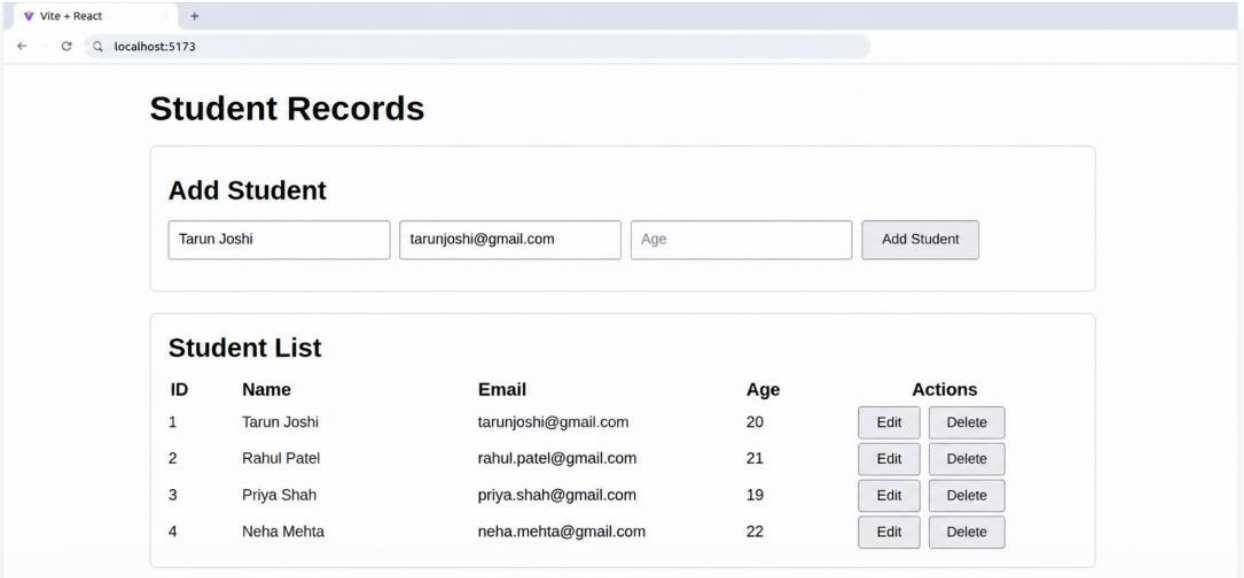
export default App;
```

Output:




Vite + React
localhost:5173
Student Records
Add Student
Tarun Joshi
tarunjoshi@gmail.com
Age
Add Student
Student added successfully
Student List
ID
Name
Email
Age
Actions
1
Tarun Joshi
tarunjoshi@gmail.com
20
Edit
Delete


Vite + React
localhost:5173
Student Records
Add Student
Student Name
Email Address
Age
Add Student
Student deleted successfully
Student List
ID
Name
Email
Age
Actions
No student records available.


Vite + React
localhost:5173
Student Records
Add Student
Tarun Joshi
tarunjoshi@gmail.com
Age
Add Student
Student List
ID
Name
Email
Age
Actions
1
Tarun Joshi
tarunjoshi@gmail.com
20
Edit
Delete
2
Rahul Patel
rahul.patel@gmail.com
21
Edit
Delete
3
Priya Shah
priya.shah@gmail.com
19
Edit
Delete
4
Neha Mehta
neha.mehta@gmail.com
22
Edit
Delete